Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 12:35:48

Measurement Details:

- **Sample:** Muestra11
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

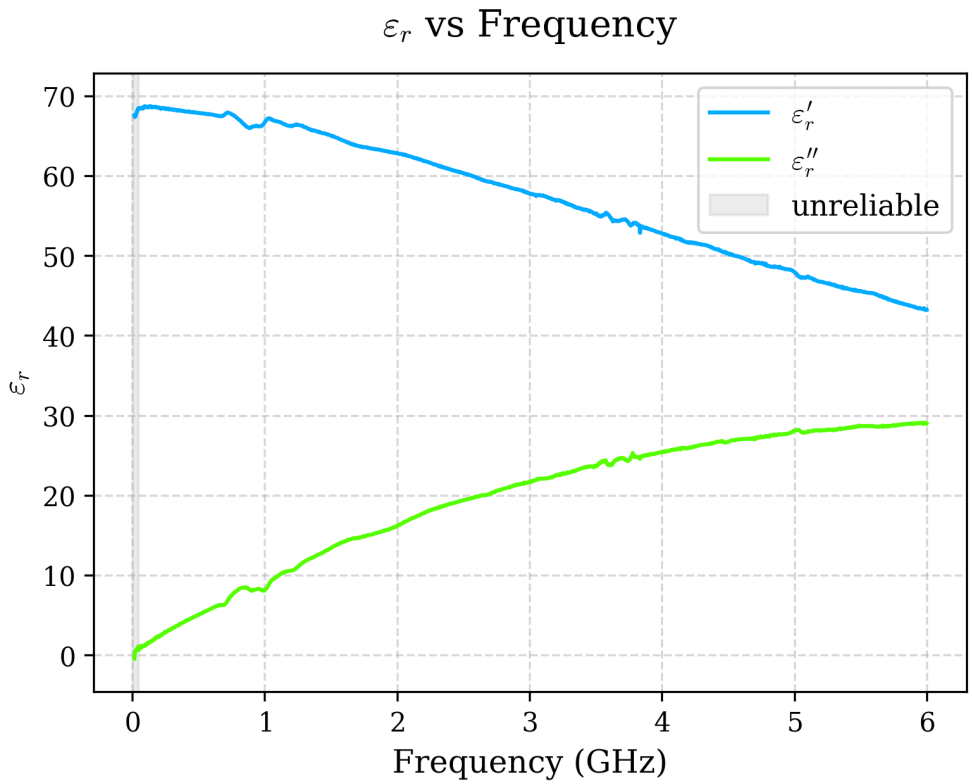
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table). Non-physical points ($\varepsilon_r'' < 0$ or $\varepsilon_r' < 1$): 1 of 1200, kept and marked † in the table, not discarded.

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	67.5525	-0.4524†	-0.0067
0.020	67.3462	0.4382	0.0065
0.025	67.5233	0.5105	0.0076
0.030	67.7078	0.6898	0.0102
0.035	68.0507	0.7883	0.0116
0.040	68.2539	0.8824	0.0129
0.045	68.4607	1.0698	0.0156
0.050	68.3677	0.6289	0.0092
0.055	68.4003	0.8907	0.0130
0.060	68.4371	0.8906	0.0130
0.065	68.5043	1.1269	0.0165
0.070	68.4963	0.9412	0.0137
0.075	68.3966	1.0224	0.0149
0.080	68.5314	1.1881	0.0173

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.085	68.4433	1.1447	0.0167
0.090	68.6817	1.1126	0.0162
0.095	68.6420	1.1593	0.0169
0.100	68.6903	1.2293	0.0179
0.105	68.5913	1.4033	0.0205
0.110	68.5864	1.3198	0.0192
0.115	68.6035	1.5062	0.0220
0.120	68.5365	1.5453	0.0225
0.125	68.6851	1.5070	0.0219
0.130	68.6618	1.6039	0.0234
0.135	68.6921	1.6041	0.0234
0.140	68.7011	1.7465	0.0254
0.145	68.5517	1.7261	0.0252
0.150	68.5599	1.7517	0.0255
0.155	68.5963	1.8439	0.0269
0.160	68.6048	1.9531	0.0285
0.165	68.6379	1.9667	0.0287
0.170	68.5964	1.9772	0.0288
0.175	68.6223	2.1876	0.0319
0.180	68.5706	2.1162	0.0309
0.185	68.5902	2.1696	0.0316
0.190	68.5213	2.3080	0.0337
0.195	68.5491	2.2539	0.0329
0.200	68.4989	2.3873	0.0349
0.205	68.5332	2.2926	0.0335
0.210	68.5058	2.4749	0.0361
0.215	68.3593	2.4899	0.0364
0.220	68.4710	2.5723	0.0376
0.225	68.4695	2.5489	0.0372
0.230	68.4206	2.5834	0.0378
0.235	68.4899	2.6671	0.0389
0.240	68.4669	2.7216	0.0398
0.245	68.4147	2.8833	0.0421
0.250	68.3413	2.9152	0.0427
0.255	68.4159	2.8643	0.0419
0.260	68.4208	2.9370	0.0429
0.265	68.3552	2.9764	0.0435
0.270	68.3275	3.0727	0.0450
0.275	68.4111	3.1324	0.0458
0.280	68.2845	3.1555	0.0462
0.285	68.3490	3.1061	0.0454
0.290	68.3041	3.2236	0.0472
0.295	68.3002	3.2787	0.0480
0.300	68.3085	3.2954	0.0482
0.305	68.2430	3.4041	0.0499
0.310	68.2235	3.4188	0.0501

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.315	68.3119	3.4485	0.0505
0.320	68.2579	3.4667	0.0508
0.325	68.2186	3.5755	0.0524
0.330	68.1916	3.6210	0.0531
0.335	68.1917	3.6490	0.0535
0.340	68.1360	3.6720	0.0539
0.345	68.2372	3.6692	0.0538
0.350	68.1840	3.8074	0.0558
0.355	68.1589	3.8652	0.0567
0.360	68.1756	3.8332	0.0562
0.365	68.1665	3.9169	0.0575
0.370	68.1012	3.9606	0.0582
0.375	68.1117	4.0324	0.0592
0.380	68.1061	4.0594	0.0596
0.385	68.1030	4.1029	0.0602
0.390	68.0994	4.1405	0.0608
0.395	68.0521	4.1751	0.0614
0.400	68.0250	4.2264	0.0621
0.405	68.0377	4.3205	0.0635
0.410	68.0356	4.2751	0.0628
0.415	68.0210	4.3423	0.0638
0.420	68.0271	4.4126	0.0649
0.425	68.0567	4.4321	0.0651
0.430	68.0289	4.4611	0.0656
0.435	67.9574	4.4901	0.0661
0.440	67.9693	4.5316	0.0667
0.445	67.9706	4.6404	0.0683
0.450	67.9581	4.6536	0.0685
0.455	67.9121	4.6566	0.0686
0.460	67.9687	4.7124	0.0693
0.465	67.9311	4.7666	0.0702
0.470	67.9354	4.8173	0.0709
0.475	67.9161	4.8546	0.0715
0.480	67.9142	4.8646	0.0716
0.485	67.8934	4.9247	0.0725
0.490	67.8678	4.9716	0.0733
0.495	67.8569	5.0189	0.0740
0.500	67.8534	5.0194	0.0740
0.505	67.8509	5.0894	0.0750
0.510	67.8497	5.1194	0.0755
0.515	67.8178	5.1762	0.0763
0.520	67.8156	5.1625	0.0761
0.525	67.7876	5.2329	0.0772
0.530	67.7960	5.2945	0.0781
0.535	67.7885	5.3090	0.0783
0.540	67.7982	5.3853	0.0794

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.545	67.8000	5.4285	0.0801
0.550	67.7617	5.4345	0.0802
0.555	67.7530	5.4794	0.0809
0.560	67.7028	5.5274	0.0816
0.565	67.6963	5.5627	0.0822
0.570	67.6992	5.6095	0.0829
0.575	67.7365	5.6368	0.0832
0.580	67.7146	5.6832	0.0839
0.585	67.6878	5.7040	0.0843
0.590	67.6706	5.7520	0.0850
0.595	67.6739	5.8128	0.0859
0.600	67.6777	5.8594	0.0866
0.605	67.6481	5.8836	0.0870
0.610	67.6092	5.9457	0.0879
0.615	67.6129	5.9784	0.0884
0.620	67.5961	5.9916	0.0886
0.625	67.5666	6.0472	0.0895
0.630	67.5485	6.0635	0.0898
0.635	67.5447	6.1382	0.0909
0.640	67.5303	6.1374	0.0909
0.645	67.4849	6.1766	0.0915
0.650	67.4852	6.2228	0.0922
0.655	67.4877	6.2401	0.0925
0.660	67.4723	6.2273	0.0923
0.665	67.4865	6.2472	0.0926
0.670	67.4404	6.2835	0.0932
0.675	67.4551	6.2320	0.0924
0.680	67.4632	6.2634	0.0928
0.685	67.4940	6.2391	0.0924
0.690	67.5655	6.2430	0.0924
0.695	67.6118	6.2823	0.0929
0.700	67.7007	6.3722	0.0941
0.705	67.7833	6.4569	0.0953
0.710	67.8356	6.5401	0.0964
0.715	67.8868	6.6795	0.0984
0.720	67.9120	6.8241	0.1005
0.725	67.8867	6.9391	0.1022
0.730	67.8687	7.0405	0.1037
0.735	67.8271	7.2046	0.1062
0.740	67.7991	7.2828	0.1074
0.745	67.7618	7.3992	0.1092
0.750	67.6887	7.5024	0.1108
0.755	67.6496	7.5768	0.1120
0.760	67.6210	7.6492	0.1131
0.765	67.5590	7.7359	0.1145
0.770	67.4943	7.8199	0.1159

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.775	67.4586	7.8605	0.1165
0.780	67.3657	7.9625	0.1182
0.785	67.2936	8.0058	0.1190
0.790	67.2439	8.0442	0.1196
0.795	67.1719	8.1097	0.1207
0.800	67.0828	8.1662	0.1217
0.805	67.0126	8.2678	0.1234
0.810	66.9664	8.2928	0.1238
0.815	66.9155	8.3507	0.1248
0.820	66.8258	8.3805	0.1254
0.825	66.7061	8.4014	0.1259
0.830	66.6704	8.4157	0.1262
0.835	66.5637	8.4554	0.1270
0.840	66.4608	8.4329	0.1269
0.845	66.4299	8.4212	0.1268
0.850	66.3239	8.4304	0.1271
0.855	66.2681	8.4706	0.1278
0.860	66.1634	8.4104	0.1271
0.865	66.1126	8.4426	0.1277
0.870	66.0888	8.4122	0.1273
0.875	66.0385	8.3014	0.1257
0.880	65.9635	8.2407	0.1249
0.885	65.9566	8.1800	0.1240
0.890	66.0127	8.1645	0.1237
0.895	66.0063	8.0808	0.1224
0.900	66.0885	8.0261	0.1214
0.905	66.1547	8.0791	0.1221
0.910	66.1671	8.1139	0.1226
0.915	66.1518	8.1331	0.1229
0.920	66.1624	8.1326	0.1229
0.925	66.2240	8.1870	0.1236
0.930	66.2342	8.1891	0.1236
0.935	66.2180	8.2084	0.1240
0.940	66.2609	8.2579	0.1246
0.945	66.2132	8.2733	0.1250
0.950	66.1867	8.2572	0.1248
0.955	66.1898	8.2870	0.1252
0.960	66.1602	8.2569	0.1248
0.965	66.1835	8.2565	0.1248
0.970	66.2063	8.2036	0.1239
0.975	66.2100	8.1369	0.1229
0.980	66.2973	8.0922	0.1221
0.985	66.3727	8.0587	0.1214
0.990	66.4497	8.0667	0.1214
0.995	66.5635	8.0469	0.1209
1.000	66.6604	8.0726	0.1211

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.005	66.8229	8.1841	0.1225
1.010	66.9406	8.2921	0.1239
1.015	66.9819	8.3804	0.1251
1.020	67.0831	8.5290	0.1271
1.025	67.1664	8.6465	0.1287
1.030	67.1597	8.8013	0.1311
1.035	67.1856	8.9171	0.1327
1.040	67.1652	9.0677	0.1350
1.045	67.1128	9.1888	0.1369
1.050	67.0809	9.3011	0.1387
1.055	66.9684	9.3713	0.1399
1.060	66.9815	9.4672	0.1413
1.065	66.9111	9.4715	0.1416
1.070	66.9087	9.5583	0.1429
1.075	66.8617	9.6427	0.1442
1.080	66.8415	9.6598	0.1445
1.085	66.8222	9.6829	0.1449
1.090	66.8212	9.7670	0.1462
1.095	66.7894	9.8482	0.1475
1.100	66.7786	9.8947	0.1482
1.105	66.7599	9.9566	0.1491
1.110	66.7266	10.0370	0.1504
1.115	66.7093	10.0920	0.1513
1.120	66.6833	10.1478	0.1522
1.125	66.6397	10.1790	0.1527
1.130	66.5829	10.2606	0.1541
1.135	66.5248	10.2990	0.1548
1.140	66.4946	10.3012	0.1549
1.145	66.4645	10.3788	0.1562
1.150	66.4374	10.3558	0.1559
1.155	66.4133	10.4279	0.1570
1.160	66.3397	10.4317	0.1572
1.165	66.3448	10.4743	0.1579
1.170	66.2983	10.4785	0.1581
1.175	66.2562	10.5114	0.1586
1.180	66.2545	10.5311	0.1589
1.185	66.2420	10.5260	0.1589
1.190	66.2155	10.5202	0.1589
1.195	66.2370	10.5549	0.1594
1.200	66.2561	10.5691	0.1595
1.205	66.1970	10.5823	0.1599
1.210	66.2550	10.5939	0.1599
1.215	66.2551	10.6343	0.1605
1.220	66.3154	10.6447	0.1605
1.225	66.3304	10.6958	0.1613
1.230	66.3698	10.7751	0.1623

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.235	66.3445	10.8237	0.1631
1.240	66.3919	10.8917	0.1641
1.245	66.4061	10.9710	0.1652
1.250	66.3532	11.0520	0.1666
1.255	66.3365	11.1288	0.1678
1.260	66.3424	11.1903	0.1687
1.265	66.3135	11.2764	0.1700
1.270	66.2939	11.3236	0.1708
1.275	66.2819	11.3872	0.1718
1.280	66.2438	11.4730	0.1732
1.285	66.2640	11.5314	0.1740
1.290	66.1470	11.5730	0.1750
1.295	66.1156	11.6423	0.1761
1.300	66.1128	11.6937	0.1769
1.305	66.0745	11.7411	0.1777
1.310	66.0452	11.7941	0.1786
1.315	66.0186	11.8389	0.1793
1.320	65.9462	11.8636	0.1799
1.325	65.9070	11.9342	0.1811
1.330	65.8808	11.9628	0.1816
1.335	65.8391	11.9959	0.1822
1.340	65.8358	12.0266	0.1827
1.345	65.7635	12.0740	0.1836
1.350	65.7382	12.1544	0.1849
1.355	65.6794	12.1291	0.1847
1.360	65.7195	12.1741	0.1852
1.365	65.6721	12.2297	0.1862
1.370	65.6476	12.2564	0.1867
1.375	65.6684	12.3039	0.1874
1.380	65.6136	12.3707	0.1885
1.385	65.6004	12.4239	0.1894
1.390	65.5715	12.4421	0.1897
1.395	65.5522	12.5071	0.1908
1.400	65.5473	12.5161	0.1909
1.405	65.4786	12.5805	0.1921
1.410	65.4919	12.6270	0.1928
1.415	65.4250	12.6332	0.1931
1.420	65.4421	12.6736	0.1937
1.425	65.4023	12.6835	0.1939
1.430	65.3561	12.7672	0.1953
1.435	65.3274	12.7762	0.1956
1.440	65.3219	12.8494	0.1967
1.445	65.3062	12.8929	0.1974
1.450	65.3166	12.9177	0.1978
1.455	65.2701	12.9986	0.1992
1.460	65.2631	13.0669	0.2002

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.465	65.2218	13.0892	0.2007
1.470	65.1844	13.1167	0.2012
1.475	65.1438	13.1064	0.2012
1.480	65.1574	13.2018	0.2026
1.485	65.1148	13.2342	0.2032
1.490	65.1115	13.2842	0.2040
1.495	65.0550	13.3316	0.2049
1.500	65.0325	13.4033	0.2061
1.505	64.9932	13.4614	0.2071
1.510	64.9250	13.4748	0.2075
1.515	64.9751	13.5524	0.2086
1.520	64.9067	13.5944	0.2094
1.525	64.8880	13.6560	0.2105
1.530	64.8478	13.6631	0.2107
1.535	64.8126	13.7096	0.2115
1.540	64.7606	13.7511	0.2123
1.545	64.7395	13.8297	0.2136
1.550	64.7307	13.8470	0.2139
1.555	64.6758	13.8517	0.2142
1.560	64.6314	13.9167	0.2153
1.565	64.5911	13.9768	0.2164
1.570	64.5434	14.0021	0.2169
1.575	64.5315	14.0538	0.2178
1.580	64.5021	14.0580	0.2179
1.585	64.4329	14.1039	0.2189
1.590	64.4073	14.1374	0.2195
1.595	64.3817	14.1683	0.2201
1.600	64.3460	14.2221	0.2210
1.605	64.3247	14.2153	0.2210
1.610	64.2629	14.2273	0.2214
1.615	64.2400	14.3009	0.2226
1.620	64.2321	14.3621	0.2236
1.625	64.2087	14.4014	0.2243
1.630	64.1214	14.3724	0.2241
1.635	64.1159	14.4222	0.2249
1.640	64.1013	14.4411	0.2253
1.645	64.0427	14.4751	0.2260
1.650	63.9731	14.4811	0.2264
1.655	63.9697	14.5043	0.2267
1.660	63.9338	14.5615	0.2278
1.665	63.9746	14.6065	0.2283
1.670	63.9008	14.6219	0.2288
1.675	63.8412	14.6258	0.2291
1.680	63.8717	14.6303	0.2291
1.685	63.7912	14.5804	0.2286
1.690	63.8037	14.6379	0.2294

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.695	63.7811	14.6294	0.2294
1.700	63.7400	14.6598	0.2300
1.705	63.7248	14.6945	0.2306
1.710	63.7043	14.6833	0.2305
1.715	63.6778	14.6452	0.2300
1.720	63.6535	14.7427	0.2316
1.725	63.6209	14.7313	0.2315
1.730	63.6721	14.7585	0.2318
1.735	63.6241	14.7892	0.2324
1.740	63.5878	14.8251	0.2331
1.745	63.5472	14.8752	0.2341
1.750	63.5735	14.8086	0.2329
1.755	63.5817	14.8811	0.2340
1.760	63.5513	14.8757	0.2341
1.765	63.5692	14.9001	0.2344
1.770	63.5281	14.9530	0.2354
1.775	63.5539	15.0021	0.2361
1.780	63.4999	14.9899	0.2361
1.785	63.4707	15.0302	0.2368
1.790	63.4405	15.0171	0.2367
1.795	63.4582	15.0362	0.2369
1.800	63.4249	15.0392	0.2371
1.805	63.3900	15.0754	0.2378
1.810	63.3902	15.0892	0.2380
1.815	63.3915	15.1086	0.2383
1.820	63.3663	15.1396	0.2389
1.825	63.3693	15.1669	0.2393
1.830	63.3113	15.2195	0.2404
1.835	63.3321	15.2430	0.2407
1.840	63.3123	15.2638	0.2411
1.845	63.2996	15.3062	0.2418
1.850	63.2893	15.2971	0.2417
1.855	63.2808	15.3216	0.2421
1.860	63.2555	15.3134	0.2421
1.865	63.2665	15.3625	0.2428
1.870	63.2533	15.3875	0.2433
1.875	63.2283	15.4250	0.2440
1.880	63.2097	15.4223	0.2440
1.885	63.1562	15.4606	0.2448
1.890	63.1588	15.4772	0.2451
1.895	63.1343	15.5194	0.2458
1.900	63.1282	15.5824	0.2468
1.905	63.1313	15.5591	0.2465
1.910	63.0966	15.5901	0.2471
1.915	63.0798	15.6329	0.2478
1.920	63.0503	15.6412	0.2481

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.925	63.0595	15.6728	0.2485
1.930	63.0270	15.6740	0.2487
1.935	63.0423	15.7225	0.2494
1.940	62.9646	15.6981	0.2493
1.945	62.9711	15.8082	0.2510
1.950	62.9338	15.8198	0.2514
1.955	62.9459	15.8619	0.2520
1.960	62.9512	15.8579	0.2519
1.965	62.8919	15.9145	0.2530
1.970	62.8932	15.9242	0.2532
1.975	62.8971	16.0038	0.2544
1.980	62.8477	16.0116	0.2548
1.985	62.8688	16.0155	0.2547
1.990	62.8563	16.0713	0.2557
1.995	62.8264	16.1337	0.2568
2.000	62.7979	16.1807	0.2577
2.005	62.8051	16.1810	0.2576
2.010	62.7891	16.2846	0.2594
2.015	62.7339	16.2690	0.2593
2.020	62.7127	16.3233	0.2603
2.025	62.7314	16.3756	0.2610
2.030	62.7345	16.3756	0.2610
2.035	62.7082	16.4561	0.2624
2.040	62.6813	16.4959	0.2632
2.045	62.5962	16.5291	0.2641
2.050	62.6747	16.6292	0.2653
2.055	62.6386	16.6394	0.2656
2.060	62.5904	16.6917	0.2667
2.065	62.6181	16.7231	0.2671
2.070	62.5297	16.7450	0.2678
2.075	62.4828	16.7888	0.2687
2.080	62.4911	16.8414	0.2695
2.085	62.4819	16.8850	0.2702
2.090	62.4466	16.9168	0.2709
2.095	62.4160	16.9628	0.2718
2.100	62.4084	17.0415	0.2731
2.105	62.3779	17.0565	0.2734
2.110	62.3761	17.1004	0.2742
2.115	62.3257	17.1307	0.2749
2.120	62.2868	17.1386	0.2752
2.125	62.2486	17.2072	0.2764
2.130	62.2253	17.2224	0.2768
2.135	62.2193	17.2729	0.2776
2.140	62.2081	17.2939	0.2780
2.145	62.1565	17.3282	0.2788
2.150	62.1754	17.3564	0.2792

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.155	62.1141	17.4054	0.2802
2.160	62.0600	17.4460	0.2811
2.165	62.0962	17.4830	0.2815
2.170	62.0332	17.5199	0.2824
2.175	62.0390	17.5696	0.2832
2.180	61.9997	17.6004	0.2839
2.185	61.9842	17.6364	0.2845
2.190	61.9342	17.6809	0.2855
2.195	61.8568	17.7327	0.2867
2.200	61.8607	17.7639	0.2872
2.205	61.8261	17.7618	0.2873
2.210	61.8346	17.8549	0.2888
2.215	61.7891	17.8591	0.2890
2.220	61.7815	17.8965	0.2897
2.225	61.7571	17.9537	0.2907
2.230	61.7109	18.0117	0.2919
2.235	61.7124	18.0080	0.2918
2.240	61.6699	18.0186	0.2922
2.245	61.6297	18.0981	0.2937
2.250	61.6063	18.1229	0.2942
2.255	61.5889	18.1253	0.2943
2.260	61.5501	18.1944	0.2956
2.265	61.5253	18.2341	0.2964
2.270	61.4749	18.2061	0.2962
2.275	61.4298	18.2580	0.2972
2.280	61.4146	18.2676	0.2974
2.285	61.4279	18.3364	0.2985
2.290	61.3704	18.3546	0.2991
2.295	61.3697	18.3296	0.2987
2.300	61.2941	18.3485	0.2994
2.305	61.2786	18.4603	0.3013
2.310	61.2351	18.3843	0.3002
2.315	61.2654	18.4487	0.3011
2.320	61.2150	18.4981	0.3022
2.325	61.1278	18.4855	0.3024
2.330	61.2112	18.5965	0.3038
2.335	61.1114	18.5914	0.3042
2.340	61.1284	18.5530	0.3035
2.345	61.0497	18.6488	0.3055
2.350	61.0432	18.6865	0.3061
2.355	60.9854	18.6453	0.3057
2.360	61.0023	18.6989	0.3065
2.365	60.9863	18.7549	0.3075
2.370	60.9588	18.7613	0.3078
2.375	60.8980	18.7440	0.3078
2.380	60.9216	18.8210	0.3089

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.385	60.8776	18.8525	0.3097
2.390	60.8644	18.8543	0.3098
2.395	60.8308	18.9381	0.3113
2.400	60.7814	18.9588	0.3119
2.405	60.7405	18.9168	0.3114
2.410	60.7519	18.9514	0.3119
2.415	60.7020	18.9950	0.3129
2.420	60.7128	19.0459	0.3137
2.425	60.6478	19.0283	0.3138
2.430	60.6174	19.0286	0.3139
2.435	60.6392	19.0532	0.3142
2.440	60.5939	19.0951	0.3151
2.445	60.5731	19.1199	0.3156
2.450	60.5745	19.1232	0.3157
2.455	60.5178	19.1448	0.3164
2.460	60.5107	19.1751	0.3169
2.465	60.5124	19.2223	0.3177
2.470	60.4424	19.2343	0.3182
2.475	60.4370	19.2469	0.3185
2.480	60.4003	19.2731	0.3191
2.485	60.4206	19.3051	0.3195
2.490	60.3711	19.3146	0.3199
2.495	60.3394	19.3725	0.3211
2.500	60.3437	19.3591	0.3208
2.505	60.2994	19.3778	0.3214
2.510	60.2721	19.3958	0.3218
2.515	60.2634	19.4227	0.3223
2.520	60.2218	19.4638	0.3232
2.525	60.1600	19.5027	0.3242
2.530	60.1555	19.5044	0.3242
2.535	60.1810	19.5174	0.3243
2.540	60.1172	19.5277	0.3248
2.545	60.0969	19.5423	0.3252
2.550	60.0633	19.5873	0.3261
2.555	60.0585	19.5913	0.3262
2.560	60.0180	19.6400	0.3272
2.565	59.9769	19.6474	0.3276
2.570	59.9530	19.6585	0.3279
2.575	59.9451	19.6863	0.3284
2.580	59.8872	19.6988	0.3289
2.585	59.8727	19.7791	0.3304
2.590	59.8232	19.7383	0.3299
2.595	59.8031	19.7981	0.3311
2.600	59.7610	19.8153	0.3316
2.605	59.6815	19.8321	0.3323
2.610	59.7305	19.8284	0.3320

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.615	59.6598	19.8313	0.3324
2.620	59.6184	19.9059	0.3339
2.625	59.5822	19.9149	0.3342
2.630	59.5831	19.9445	0.3347
2.635	59.5617	19.9311	0.3346
2.640	59.4806	19.9795	0.3359
2.645	59.4764	19.9753	0.3359
2.650	59.4186	20.0276	0.3371
2.655	59.4159	20.0048	0.3367
2.660	59.3631	20.0250	0.3373
2.665	59.3807	20.0101	0.3370
2.670	59.3196	19.9906	0.3370
2.675	59.3168	20.0418	0.3379
2.680	59.2929	20.0617	0.3383
2.685	59.2646	20.0900	0.3390
2.690	59.2617	20.1092	0.3393
2.695	59.2387	20.1441	0.3400
2.700	59.2380	20.1836	0.3407
2.705	59.2191	20.2113	0.3413
2.710	59.2458	20.2769	0.3422
2.715	59.2044	20.2520	0.3421
2.720	59.1626	20.3254	0.3436
2.725	59.0987	20.3624	0.3445
2.730	59.0585	20.4131	0.3456
2.735	59.0731	20.4590	0.3463
2.740	59.0221	20.3944	0.3455
2.745	59.0335	20.5117	0.3475
2.750	59.0100	20.5443	0.3481
2.755	58.9772	20.5746	0.3489
2.760	58.9911	20.5793	0.3489
2.765	58.9152	20.5924	0.3495
2.770	58.9519	20.6550	0.3504
2.775	58.8691	20.6489	0.3508
2.780	58.8549	20.6692	0.3512
2.785	58.8624	20.7100	0.3518
2.790	58.8269	20.6973	0.3518
2.795	58.8145	20.7623	0.3530
2.800	58.7800	20.7808	0.3535
2.805	58.7722	20.8087	0.3541
2.810	58.7144	20.8473	0.3551
2.815	58.7193	20.8678	0.3554
2.820	58.7070	20.8629	0.3554
2.825	58.6619	20.8880	0.3561
2.830	58.6435	20.9298	0.3569
2.835	58.6373	20.9400	0.3571
2.840	58.5948	21.0152	0.3587

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.845	58.5818	21.0222	0.3589
2.850	58.5238	20.9759	0.3584
2.855	58.5346	21.0445	0.3595
2.860	58.4706	21.0584	0.3602
2.865	58.4455	21.1275	0.3615
2.870	58.4358	21.1256	0.3615
2.875	58.4094	21.1371	0.3619
2.880	58.4118	21.2000	0.3629
2.885	58.3731	21.1977	0.3631
2.890	58.3574	21.3010	0.3650
2.895	58.3013	21.2525	0.3645
2.900	58.3221	21.2587	0.3645
2.905	58.2961	21.3193	0.3657
2.910	58.3046	21.3595	0.3663
2.915	58.2693	21.3444	0.3663
2.920	58.2470	21.3721	0.3669
2.925	58.2161	21.3297	0.3664
2.930	58.1997	21.3687	0.3672
2.935	58.1618	21.4089	0.3681
2.940	58.1653	21.4171	0.3682
2.945	58.1252	21.4505	0.3690
2.950	58.0774	21.4737	0.3697
2.955	58.1277	21.4203	0.3685
2.960	57.9899	21.6013	0.3725
2.965	58.0094	21.4647	0.3700
2.970	57.9248	21.5045	0.3712
2.975	57.9121	21.5387	0.3719
2.980	57.9549	21.5531	0.3719
2.985	57.8457	21.5986	0.3734
2.990	57.8588	21.6534	0.3742
2.995	57.8174	21.5953	0.3735
3.000	57.7590	21.7122	0.3759
3.005	57.7667	21.6671	0.3751
3.010	57.7174	21.6384	0.3749
3.015	57.6848	21.6862	0.3759
3.020	57.7114	21.8009	0.3778
3.025	57.6608	21.7823	0.3778
3.030	57.6468	21.7630	0.3775
3.035	57.6572	21.8409	0.3788
3.040	57.6100	21.8971	0.3801
3.045	57.5535	21.8927	0.3804
3.050	57.4591	21.9152	0.3814
3.055	57.5839	22.0017	0.3821
3.060	57.5829	22.0312	0.3826
3.065	57.5937	21.9543	0.3812
3.070	57.5686	22.0861	0.3836

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.075	57.5812	22.0636	0.3832
3.080	57.5773	22.0859	0.3836
3.085	57.5047	22.0762	0.3839
3.090	57.5339	22.0872	0.3839
3.095	57.5355	22.0928	0.3840
3.100	57.5480	22.0700	0.3835
3.105	57.4531	22.0663	0.3841
3.110	57.4719	22.0953	0.3845
3.115	57.4472	22.0854	0.3844
3.120	57.4322	22.1127	0.3850
3.125	57.3653	22.0968	0.3852
3.130	57.3542	22.1234	0.3857
3.135	57.3033	22.1421	0.3864
3.140	57.3042	22.1530	0.3866
3.145	57.2217	22.1578	0.3872
3.150	57.2324	22.1755	0.3875
3.155	57.1848	22.2271	0.3887
3.160	57.1682	22.2031	0.3884
3.165	57.1062	22.2730	0.3900
3.170	57.1150	22.2246	0.3891
3.175	57.0427	22.2824	0.3906
3.180	57.0237	22.3308	0.3916
3.185	56.9851	22.3900	0.3929
3.190	56.9807	22.3850	0.3929
3.195	56.9611	22.4798	0.3947
3.200	56.8626	22.4782	0.3953
3.205	56.9471	22.5420	0.3958
3.210	56.9103	22.5494	0.3962
3.215	56.9189	22.5423	0.3960
3.220	56.9448	22.5454	0.3959
3.225	56.9091	22.6268	0.3976
3.230	56.8645	22.5056	0.3958
3.235	56.8793	22.6112	0.3975
3.240	56.7870	22.6542	0.3989
3.245	56.7884	22.5899	0.3978
3.250	56.7574	22.7612	0.4010
3.255	56.6607	22.7026	0.4007
3.260	56.7139	22.7061	0.4004
3.265	56.6816	22.7669	0.4017
3.270	56.5616	22.6867	0.4011
3.275	56.5469	22.7391	0.4021
3.280	56.5668	22.7385	0.4020
3.285	56.4337	22.8827	0.4055
3.290	56.4857	22.7804	0.4033
3.295	56.3928	22.8187	0.4046
3.300	56.4272	22.8066	0.4042

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.305	56.3863	22.8967	0.4061
3.310	56.3451	22.8866	0.4062
3.315	56.2777	22.9143	0.4072
3.320	56.3079	22.9691	0.4079
3.325	56.2800	22.9236	0.4073
3.330	56.2239	23.0270	0.4096
3.335	56.1028	23.0328	0.4105
3.340	56.1536	23.0560	0.4106
3.345	56.0536	23.0255	0.4108
3.350	56.0262	23.1107	0.4125
3.355	56.0430	23.1026	0.4122
3.360	55.9704	23.1972	0.4145
3.365	55.9879	23.2081	0.4145
3.370	55.9196	23.2434	0.4157
3.375	55.9685	23.2722	0.4158
3.380	55.9316	23.2777	0.4162
3.385	55.9001	23.3112	0.4170
3.390	55.8384	23.3411	0.4180
3.395	55.8519	23.3437	0.4180
3.400	55.7903	23.4359	0.4201
3.405	55.7480	23.4284	0.4203
3.410	55.8051	23.4411	0.4201
3.415	55.6879	23.5140	0.4222
3.420	55.7107	23.4782	0.4214
3.425	55.7511	23.4701	0.4210
3.430	55.7144	23.5107	0.4220
3.435	55.7190	23.5653	0.4229
3.440	55.6497	23.5125	0.4225
3.445	55.6288	23.5766	0.4238
3.450	55.5177	23.6158	0.4254
3.455	55.6273	23.6184	0.4246
3.460	55.5105	23.6441	0.4259
3.465	55.4918	23.6177	0.4256
3.470	55.4754	23.6042	0.4255
3.475	55.4331	23.5743	0.4253
3.480	55.3584	23.5650	0.4257
3.485	55.2676	23.4962	0.4251
3.490	55.2930	23.7131	0.4289
3.495	55.2575	23.7026	0.4289
3.500	55.1856	23.6736	0.4290
3.505	55.2848	23.6746	0.4282
3.510	55.1050	23.6502	0.4292
3.515	55.0990	23.7862	0.4317
3.520	55.0057	23.8115	0.4329
3.525	54.9165	24.0110	0.4372
3.530	55.0048	24.0564	0.4374

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.535	54.9151	24.0177	0.4374
3.540	54.8797	24.1504	0.4401
3.545	55.0246	24.2075	0.4399
3.550	54.9612	24.2977	0.4421
3.555	55.0288	24.2185	0.4401
3.560	55.1188	24.2871	0.4406
3.565	55.1422	24.2188	0.4392
3.570	55.1113	24.3317	0.4415
3.575	55.3359	24.3847	0.4407
3.580	55.3579	24.2732	0.4385
3.585	55.2966	24.0058	0.4341
3.590	55.2239	23.8373	0.4316
3.595	55.1257	23.7738	0.4313
3.600	55.0533	23.8729	0.4336
3.605	54.9261	23.7370	0.4322
3.610	54.8277	23.7533	0.4332
3.615	54.6917	23.8605	0.4363
3.620	54.5360	23.9118	0.4385
3.625	54.5266	23.8041	0.4366
3.630	54.2730	24.0608	0.4433
3.635	54.3675	24.1531	0.4443
3.640	54.4345	24.2056	0.4447
3.645	54.4054	24.3241	0.4471
3.650	54.3521	24.4127	0.4492
3.655	54.3154	24.3848	0.4489
3.660	54.3773	24.5168	0.4509
3.665	54.3337	24.4471	0.4499
3.670	54.3440	24.5562	0.4519
3.675	54.3154	24.5160	0.4514
3.680	54.3630	24.5772	0.4521
3.685	54.4093	24.6635	0.4533
3.690	54.3975	24.6404	0.4530
3.695	54.4625	24.6783	0.4531
3.700	54.5067	24.6643	0.4525
3.705	54.4999	24.5838	0.4511
3.710	54.4727	24.6373	0.4523
3.715	54.5339	24.5812	0.4508
3.720	54.5511	24.4703	0.4486
3.725	54.5188	24.5186	0.4497
3.730	54.4741	24.3846	0.4476
3.735	54.3482	24.3247	0.4476
3.740	54.2935	24.2835	0.4473
3.745	54.2254	24.2856	0.4479
3.750	54.0455	24.3350	0.4503
3.755	53.9273	24.4230	0.4529
3.760	53.8288	24.5499	0.4561

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.765	53.7465	24.7429	0.4604
3.770	53.8822	24.8734	0.4616
3.775	53.9330	24.8505	0.4608
3.780	53.9115	25.2995	0.4693
3.785	54.0988	25.0135	0.4624
3.790	54.0876	24.8026	0.4586
3.795	54.0743	24.7947	0.4585
3.800	54.1119	24.8630	0.4595
3.805	54.0878	24.7610	0.4578
3.810	53.9753	24.7755	0.4590
3.815	53.8615	24.7252	0.4591
3.820	53.9200	24.7620	0.4592
3.825	53.8009	24.7362	0.4598
3.830	53.8045	24.7858	0.4607
3.835	52.8382	24.5463	0.4646
3.840	53.6747	24.8359	0.4627
3.845	53.6685	24.8815	0.4636
3.850	53.6271	24.8745	0.4638
3.855	53.6112	24.9417	0.4652
3.860	53.5330	24.9149	0.4654
3.865	53.4635	24.9985	0.4676
3.870	53.5159	24.9345	0.4659
3.875	53.4625	24.9893	0.4674
3.880	53.4764	24.9784	0.4671
3.885	53.3962	25.0530	0.4692
3.890	53.4508	25.0548	0.4687
3.895	53.3890	24.9922	0.4681
3.900	53.3235	25.0367	0.4695
3.905	53.3209	25.1435	0.4715
3.910	53.2962	25.0869	0.4707
3.915	53.2584	25.0986	0.4713
3.920	53.2585	25.1745	0.4727
3.925	53.1962	25.1325	0.4724
3.930	53.1776	25.1492	0.4729
3.935	53.1754	25.1452	0.4729
3.940	53.1361	25.2010	0.4743
3.945	53.1147	25.1588	0.4737
3.950	53.0460	25.2265	0.4756
3.955	53.0186	25.2496	0.4762
3.960	52.9883	25.2545	0.4766
3.965	52.9733	25.2632	0.4769
3.970	52.9599	25.3273	0.4782
3.975	52.8978	25.3299	0.4788
3.980	52.8852	25.3162	0.4787
3.985	52.8648	25.3647	0.4798
3.990	52.8601	25.3588	0.4797

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.995	52.8034	25.3693	0.4804
4.000	52.7766	25.4170	0.4816
4.005	52.7521	25.4133	0.4817
4.010	52.6867	25.4305	0.4827
4.015	52.7020	25.4946	0.4837
4.020	52.6484	25.4590	0.4836
4.025	52.6648	25.3976	0.4822
4.030	52.5729	25.5031	0.4851
4.035	52.5663	25.5420	0.4859
4.040	52.5789	25.5210	0.4854
4.045	52.5444	25.5028	0.4854
4.050	52.4952	25.5090	0.4859
4.055	52.4250	25.5510	0.4874
4.060	52.4065	25.6023	0.4885
4.065	52.3614	25.6458	0.4898
4.070	52.3936	25.6349	0.4893
4.075	52.3412	25.6398	0.4899
4.080	52.3307	25.7031	0.4912
4.085	52.3041	25.6842	0.4911
4.090	52.3002	25.6863	0.4911
4.095	52.2703	25.7457	0.4925
4.100	52.2743	25.7611	0.4928
4.105	52.1176	25.6820	0.4928
4.110	52.2192	25.7412	0.4929
4.115	52.1449	25.7941	0.4947
4.120	52.1275	25.8421	0.4957
4.125	52.1582	25.7664	0.4940
4.130	52.1606	25.8398	0.4954
4.135	52.0083	25.8650	0.4973
4.140	51.9747	25.8301	0.4970
4.145	52.0596	25.7424	0.4945
4.150	51.9045	25.9255	0.4995
4.155	51.8634	25.9612	0.5006
4.160	51.8927	25.9182	0.4995
4.165	51.7996	25.9148	0.5003
4.170	51.7400	25.9303	0.5012
4.175	51.7433	25.8460	0.4995
4.180	51.7438	25.8924	0.5004
4.185	51.6959	25.8513	0.5001
4.190	51.6354	25.8782	0.5012
4.195	51.6277	25.9136	0.5019
4.200	51.6680	25.8980	0.5012
4.205	51.6752	25.9079	0.5014
4.210	51.5679	25.9961	0.5041
4.215	51.6407	25.9747	0.5030
4.220	51.6161	25.9919	0.5036

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.225	51.6701	25.9645	0.5025
4.230	51.6145	26.0303	0.5043
4.235	51.5482	26.0574	0.5055
4.240	51.6195	26.1324	0.5062
4.245	51.6160	26.0423	0.5045
4.250	51.5654	26.0890	0.5059
4.255	51.5832	26.1278	0.5065
4.260	51.4970	26.1648	0.5081
4.265	51.5420	26.1324	0.5070
4.270	51.4901	26.1876	0.5086
4.275	51.4718	26.2083	0.5092
4.280	51.4569	26.2157	0.5095
4.285	51.4517	26.2808	0.5108
4.290	51.4139	26.2760	0.5111
4.295	51.3761	26.2995	0.5119
4.300	51.4067	26.3155	0.5119
4.305	51.3792	26.3182	0.5122
4.310	51.3680	26.3290	0.5126
4.315	51.3539	26.3460	0.5130
4.320	51.3405	26.3434	0.5131
4.325	51.3238	26.3814	0.5140
4.330	51.3168	26.3664	0.5138
4.335	51.3002	26.3821	0.5143
4.340	51.2611	26.3654	0.5143
4.345	51.2313	26.3855	0.5150
4.350	51.2157	26.3901	0.5153
4.355	51.1928	26.4351	0.5164
4.360	51.1354	26.4283	0.5168
4.365	51.0866	26.4580	0.5179
4.370	51.1049	26.4345	0.5173
4.375	51.0398	26.5312	0.5198
4.380	51.0138	26.4932	0.5193
4.385	50.9161	26.5546	0.5215
4.390	50.9220	26.5125	0.5206
4.395	50.8760	26.5699	0.5222
4.400	50.8531	26.6118	0.5233
4.405	50.7876	26.6383	0.5245
4.410	50.8111	26.6602	0.5247
4.415	50.7656	26.6264	0.5245
4.420	50.7791	26.7231	0.5263
4.425	50.8101	26.7064	0.5256
4.430	50.7020	26.7007	0.5266
4.435	50.6377	26.7095	0.5275
4.440	50.6727	26.7121	0.5272
4.445	50.5764	26.7613	0.5291
4.450	50.5561	26.7079	0.5283

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.455	50.5032	26.8030	0.5307
4.460	50.4680	26.6775	0.5286
4.465	50.4013	26.6306	0.5284
4.470	50.4446	26.6358	0.5280
4.475	50.4251	26.5849	0.5272
4.480	50.4700	26.5815	0.5267
4.485	50.2910	26.6210	0.5293
4.490	50.3518	26.6509	0.5293
4.495	50.3029	26.6430	0.5297
4.500	50.2832	26.6694	0.5304
4.505	50.2171	26.7097	0.5319
4.510	50.1137	26.7589	0.5340
4.515	50.2139	26.8301	0.5343
4.520	50.1417	26.8548	0.5356
4.525	50.1903	26.8290	0.5345
4.530	50.0709	26.8533	0.5363
4.535	50.1146	26.8714	0.5362
4.540	50.0615	26.8764	0.5369
4.545	50.0067	26.9122	0.5382
4.550	49.9880	26.8635	0.5374
4.555	49.9572	26.9215	0.5389
4.560	49.9591	26.9413	0.5393
4.565	49.9447	26.9807	0.5402
4.570	49.9056	26.9313	0.5396
4.575	49.8943	26.9578	0.5403
4.580	49.8897	26.9775	0.5407
4.585	49.8696	26.9799	0.5410
4.590	49.7945	26.9862	0.5420
4.595	49.8023	27.0374	0.5429
4.600	49.7775	27.0286	0.5430
4.605	49.7209	27.0397	0.5438
4.610	49.7018	27.0361	0.5440
4.615	49.6703	27.0174	0.5439
4.620	49.6429	27.0285	0.5445
4.625	49.6138	27.0502	0.5452
4.630	49.5902	27.0371	0.5452
4.635	49.5291	27.0444	0.5460
4.640	49.4904	27.0633	0.5468
4.645	49.4887	27.0628	0.5468
4.650	49.4649	27.0817	0.5475
4.655	49.3568	27.0518	0.5481
4.660	49.4253	27.0545	0.5474
4.665	49.3828	27.0527	0.5478
4.670	49.2606	27.1205	0.5506
4.675	49.2058	27.0511	0.5498
4.680	49.2246	27.0559	0.5496

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.685	49.2065	27.0584	0.5499
4.690	49.2301	27.1516	0.5515
4.695	49.1794	27.0370	0.5498
4.700	49.1795	27.1299	0.5517
4.705	48.9934	27.0072	0.5512
4.710	49.0404	27.2453	0.5556
4.715	49.0984	27.2124	0.5542
4.720	49.0434	27.1709	0.5540
4.725	49.1300	27.2842	0.5553
4.730	49.1067	27.1882	0.5537
4.735	49.1095	27.2729	0.5553
4.740	49.0844	27.2614	0.5554
4.745	49.0039	27.3722	0.5586
4.750	49.0701	27.2384	0.5551
4.755	49.0830	27.3038	0.5563
4.760	48.9834	27.2829	0.5570
4.765	49.0469	27.2504	0.5556
4.770	49.0232	27.3278	0.5574
4.775	48.8216	27.3054	0.5593
4.780	48.8329	27.3029	0.5591
4.785	48.9939	27.4652	0.5606
4.790	48.7821	27.3298	0.5602
4.795	48.6931	27.3764	0.5622
4.800	48.7120	27.3958	0.5624
4.805	48.7030	27.3038	0.5606
4.810	48.7159	27.4591	0.5637
4.815	48.6382	27.3573	0.5625
4.820	48.6133	27.5172	0.5660
4.825	48.6029	27.4818	0.5654
4.830	48.5891	27.5513	0.5670
4.835	48.6139	27.5613	0.5669
4.840	48.6138	27.5362	0.5664
4.845	48.6282	27.4633	0.5648
4.850	48.5471	27.5908	0.5683
4.855	48.6514	27.5346	0.5660
4.860	48.5932	27.5489	0.5669
4.865	48.5883	27.4964	0.5659
4.870	48.5415	27.5650	0.5679
4.875	48.5711	27.5827	0.5679
4.880	48.5682	27.6207	0.5687
4.885	48.4835	27.5915	0.5691
4.890	48.6144	27.6296	0.5683
4.895	48.4638	27.6041	0.5696
4.900	48.4898	27.6176	0.5696
4.905	48.4621	27.6212	0.5700
4.910	48.3956	27.6229	0.5708

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.915	48.3896	27.6587	0.5716
4.920	48.3780	27.6493	0.5715
4.925	48.3592	27.6415	0.5716
4.930	48.3018	27.6633	0.5727
4.935	48.3034	27.6667	0.5728
4.940	48.2993	27.6725	0.5729
4.945	48.2750	27.7054	0.5739
4.950	48.2863	27.7025	0.5737
4.955	48.2917	27.7390	0.5744
4.960	48.2359	27.7403	0.5751
4.965	48.2433	27.7771	0.5758
4.970	48.2317	27.8173	0.5767
4.975	48.2361	27.8181	0.5767
4.980	48.1742	27.9064	0.5793
4.985	48.1231	27.9251	0.5803
4.990	48.0998	27.9778	0.5817
4.995	48.0152	28.0597	0.5844
5.000	47.9578	28.0624	0.5851
5.005	47.9000	28.1070	0.5868
5.010	47.7992	28.1172	0.5882
5.015	47.7771	28.1613	0.5894
5.020	47.6491	28.1557	0.5909
5.025	47.5254	28.1868	0.5931
5.030	47.4162	28.1659	0.5940
5.035	47.4372	28.1353	0.5931
5.040	47.3289	28.1138	0.5940
5.045	47.2950	28.0112	0.5923
5.050	47.2096	27.9388	0.5918
5.055	47.2798	27.9041	0.5902
5.060	47.2246	27.8998	0.5908
5.065	47.2258	27.7960	0.5886
5.070	47.2849	27.8550	0.5891
5.075	47.2496	27.8372	0.5892
5.080	47.2140	27.8524	0.5899
5.085	47.2022	27.8453	0.5899
5.090	47.2735	27.8507	0.5891
5.095	47.3451	27.8964	0.5892
5.100	47.3072	27.9083	0.5899
5.105	47.3862	28.0300	0.5915
5.110	47.2524	27.9607	0.5917
5.115	47.2105	27.9727	0.5925
5.120	47.2167	27.9936	0.5929
5.125	47.2080	28.0096	0.5933
5.130	47.1693	28.0369	0.5944
5.135	47.1174	28.0502	0.5953
5.140	47.1091	28.1011	0.5965

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.145	46.9783	28.0308	0.5967
5.150	47.0319	28.1348	0.5982
5.155	46.9584	28.1084	0.5986
5.160	46.9303	28.1156	0.5991
5.165	46.8843	28.1265	0.5999
5.170	46.8360	28.1525	0.6011
5.175	46.8039	28.1273	0.6010
5.180	46.8012	28.1192	0.6008
5.185	46.7550	28.1385	0.6018
5.190	46.7467	28.0793	0.6007
5.195	46.7277	28.1201	0.6018
5.200	46.7160	28.1044	0.6016
5.205	46.7350	28.1238	0.6018
5.210	46.7106	28.1127	0.6018
5.215	46.6926	28.1356	0.6026
5.220	46.6401	28.1211	0.6029
5.225	46.6084	28.1037	0.6030
5.230	46.6067	28.1148	0.6032
5.235	46.5905	28.1656	0.6045
5.240	46.5508	28.1578	0.6049
5.245	46.5458	28.1663	0.6051
5.250	46.5368	28.2111	0.6062
5.255	46.4862	28.2373	0.6074
5.260	46.5241	28.1488	0.6050
5.265	46.5457	28.2012	0.6059
5.270	46.4500	28.2073	0.6073
5.275	46.3899	28.1916	0.6077
5.280	46.4223	28.2317	0.6081
5.285	46.4206	28.1897	0.6073
5.290	46.3269	28.1925	0.6086
5.295	46.3408	28.1063	0.6065
5.300	46.3151	28.1947	0.6088
5.305	46.3491	28.1585	0.6075
5.310	46.2040	28.2704	0.6119
5.315	46.2694	28.2308	0.6101
5.320	46.1561	28.1978	0.6109
5.325	46.1847	28.2186	0.6110
5.330	46.1822	28.2592	0.6119
5.335	46.0872	28.3386	0.6149
5.340	46.1042	28.3785	0.6155
5.345	46.0794	28.3926	0.6162
5.350	46.1092	28.2705	0.6131
5.355	46.0266	28.3616	0.6162
5.360	45.9501	28.2811	0.6155
5.365	46.0996	28.2916	0.6137
5.370	45.9132	28.3225	0.6169

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.375	45.9797	28.3332	0.6162
5.380	45.9606	28.3354	0.6165
5.385	45.8589	28.3503	0.6182
5.390	45.9119	28.4421	0.6195
5.395	45.9505	28.4086	0.6182
5.400	45.8498	28.4326	0.6201
5.405	45.8661	28.4264	0.6198
5.410	45.8332	28.4736	0.6212
5.415	45.7817	28.4235	0.6208
5.420	45.7832	28.4970	0.6224
5.425	45.7581	28.4898	0.6226
5.430	45.7372	28.4458	0.6219
5.435	45.7433	28.4606	0.6222
5.440	45.6691	28.5683	0.6256
5.445	45.7283	28.5134	0.6235
5.450	45.6378	28.5587	0.6258
5.455	45.7186	28.4956	0.6233
5.460	45.6269	28.5597	0.6259
5.465	45.7104	28.5737	0.6251
5.470	45.5758	28.6355	0.6283
5.475	45.5768	28.5842	0.6272
5.480	45.5823	28.6481	0.6285
5.485	45.5573	28.6117	0.6280
5.490	45.5354	28.7184	0.6307
5.495	45.5634	28.6135	0.6280
5.500	45.5498	28.6806	0.6297
5.505	45.5530	28.6696	0.6294
5.510	45.5158	28.6642	0.6298
5.515	45.4859	28.6403	0.6297
5.520	45.4492	28.6562	0.6305
5.525	45.4797	28.6333	0.6296
5.530	45.4429	28.6580	0.6306
5.535	45.4239	28.6385	0.6305
5.540	45.4132	28.6797	0.6315
5.545	45.3658	28.6699	0.6320
5.550	45.3615	28.6797	0.6322
5.555	45.3467	28.6361	0.6315
5.560	45.3390	28.6744	0.6324
5.565	45.3380	28.6577	0.6321
5.570	45.3234	28.6608	0.6324
5.575	45.3286	28.6516	0.6321
5.580	45.3275	28.6899	0.6329
5.585	45.3011	28.6723	0.6329
5.590	45.3057	28.6438	0.6322
5.595	45.2765	28.6521	0.6328
5.600	45.2775	28.6523	0.6328

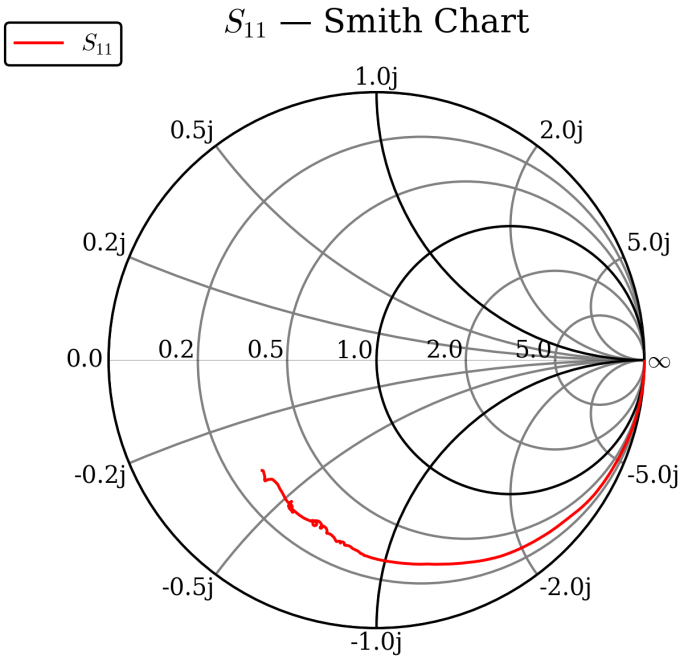
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.605	45.2658	28.6021	0.6319
5.610	45.1847	28.6030	0.6330
5.615	45.1603	28.5845	0.6330
5.620	45.1229	28.5874	0.6335
5.625	45.1336	28.6210	0.6341
5.630	45.0488	28.6076	0.6350
5.635	45.0212	28.5834	0.6349
5.640	44.9676	28.5633	0.6352
5.645	44.9204	28.5896	0.6365
5.650	44.9505	28.6173	0.6366
5.655	44.8962	28.6204	0.6375
5.660	44.8749	28.6453	0.6383
5.665	44.9215	28.5707	0.6360
5.670	44.7602	28.7270	0.6418
5.675	44.7711	28.6050	0.6389
5.680	44.7098	28.6673	0.6412
5.685	44.7723	28.5814	0.6384
5.690	44.7052	28.6272	0.6404
5.695	44.7112	28.6785	0.6414
5.700	44.5963	28.6236	0.6418
5.705	44.5244	28.6928	0.6444
5.710	44.5950	28.6287	0.6420
5.715	44.4823	28.6017	0.6430
5.720	44.4838	28.6816	0.6448
5.725	44.4506	28.6999	0.6457
5.730	44.4417	28.6731	0.6452
5.735	44.3943	28.6751	0.6459
5.740	44.3786	28.7144	0.6470
5.745	44.3585	28.7806	0.6488
5.750	44.3081	28.7460	0.6488
5.755	44.3178	28.7202	0.6480
5.760	44.2168	28.7602	0.6504
5.765	44.2696	28.6895	0.6481
5.770	44.2178	28.7270	0.6497
5.775	44.2490	28.7111	0.6489
5.780	44.1488	28.8124	0.6526
5.785	44.1410	28.7752	0.6519
5.790	44.0776	28.8257	0.6540
5.795	44.1029	28.8487	0.6541
5.800	44.0548	28.8311	0.6544
5.805	44.0251	28.8396	0.6551
5.810	44.0608	28.8526	0.6548
5.815	44.0073	28.8850	0.6564
5.820	43.9792	28.8435	0.6558
5.825	43.9583	28.9054	0.6576
5.830	43.9047	28.8860	0.6579

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.835	43.9198	28.8998	0.6580
5.840	43.8980	28.8882	0.6581
5.845	43.8572	28.9290	0.6596
5.850	43.8351	28.9246	0.6599
5.855	43.8075	28.9147	0.6600
5.860	43.7909	28.9362	0.6608
5.865	43.8052	28.9513	0.6609
5.870	43.7334	28.9160	0.6612
5.875	43.7410	28.9623	0.6621
5.880	43.7064	28.9522	0.6624
5.885	43.6930	28.9060	0.6616
5.890	43.6548	29.0244	0.6649
5.895	43.6421	28.9787	0.6640
5.900	43.6147	28.9248	0.6632
5.905	43.6154	29.0024	0.6650
5.910	43.5125	28.9671	0.6657
5.915	43.5389	29.0047	0.6662
5.920	43.5123	28.9738	0.6659
5.925	43.4562	28.9926	0.6672
5.930	43.4203	29.0485	0.6690
5.935	43.3957	29.0340	0.6691
5.940	43.3832	29.0276	0.6691
5.945	43.4628	28.9600	0.6663
5.950	43.3970	29.0651	0.6697
5.955	43.4000	29.0579	0.6695
5.960	43.3624	29.0540	0.6700
5.965	43.3287	29.0448	0.6703
5.970	43.3478	29.0955	0.6712
5.975	43.2981	28.9196	0.6679
5.980	43.4172	28.9733	0.6673
5.985	43.2795	29.0956	0.6723
5.990	43.1842	29.0342	0.6723
5.995	43.1801	29.0237	0.6722
6.000	43.2075	28.9677	0.6704

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0044	0.00
0.010	0.9998	-0.0087	-0.00
0.015	0.9999	-0.0131	-0.00
0.020	0.9997	-0.0175	-0.00
0.025	0.9996	-0.0218	-0.00
0.030	0.9995	-0.0263	-0.00
0.035	0.9992	-0.0305	-0.00
0.040	0.9990	-0.0351	-0.00
0.045	0.9988	-0.0393	-0.00
0.050	0.9986	-0.0437	-0.00
0.055	0.9983	-0.0481	-0.00
0.060	0.9980	-0.0524	-0.01
0.065	0.9977	-0.0569	-0.01
0.070	0.9974	-0.0612	-0.01
0.075	0.9970	-0.0654	-0.01
0.080	0.9966	-0.0698	-0.01
0.085	0.9962	-0.0741	-0.01
0.090	0.9957	-0.0786	-0.01
0.095	0.9951	-0.0827	-0.01
0.100	0.9947	-0.0872	-0.01
0.105	0.9942	-0.0915	-0.01
0.110	0.9937	-0.0957	-0.02
0.115	0.9930	-0.1001	-0.02

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.120	0.9923	-0.1043	-0.02
0.125	0.9918	-0.1087	-0.02
0.130	0.9911	-0.1130	-0.02
0.135	0.9905	-0.1173	-0.02
0.140	0.9898	-0.1215	-0.02
0.145	0.9891	-0.1258	-0.03
0.150	0.9884	-0.1300	-0.03
0.155	0.9875	-0.1342	-0.03
0.160	0.9867	-0.1385	-0.03
0.165	0.9860	-0.1428	-0.03
0.170	0.9852	-0.1470	-0.03
0.175	0.9843	-0.1512	-0.04
0.180	0.9835	-0.1554	-0.04
0.185	0.9825	-0.1596	-0.04
0.190	0.9814	-0.1639	-0.04
0.195	0.9806	-0.1682	-0.04
0.200	0.9795	-0.1723	-0.05
0.205	0.9786	-0.1765	-0.05
0.210	0.9775	-0.1807	-0.05
0.215	0.9764	-0.1847	-0.05
0.220	0.9754	-0.1891	-0.06
0.225	0.9743	-0.1933	-0.06
0.230	0.9731	-0.1973	-0.06
0.235	0.9718	-0.2016	-0.07
0.240	0.9708	-0.2058	-0.07
0.245	0.9695	-0.2098	-0.07
0.250	0.9683	-0.2139	-0.07
0.255	0.9670	-0.2181	-0.08
0.260	0.9657	-0.2222	-0.08
0.265	0.9645	-0.2263	-0.08
0.270	0.9631	-0.2304	-0.09
0.275	0.9617	-0.2345	-0.09
0.280	0.9604	-0.2385	-0.09
0.285	0.9591	-0.2426	-0.09
0.290	0.9576	-0.2466	-0.10
0.295	0.9561	-0.2506	-0.10
0.300	0.9548	-0.2547	-0.10
0.305	0.9532	-0.2586	-0.11
0.310	0.9518	-0.2626	-0.11
0.315	0.9501	-0.2666	-0.12
0.320	0.9486	-0.2705	-0.12
0.325	0.9471	-0.2744	-0.12
0.330	0.9454	-0.2783	-0.13
0.335	0.9438	-0.2822	-0.13
0.340	0.9423	-0.2860	-0.13
0.345	0.9406	-0.2901	-0.14

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9388	-0.2938	-0.14
0.355	0.9373	-0.2977	-0.15
0.360	0.9357	-0.3016	-0.15
0.365	0.9338	-0.3054	-0.15
0.370	0.9320	-0.3091	-0.16
0.375	0.9302	-0.3130	-0.16
0.380	0.9284	-0.3167	-0.17
0.385	0.9267	-0.3205	-0.17
0.390	0.9250	-0.3242	-0.17
0.395	0.9231	-0.3279	-0.18
0.400	0.9214	-0.3316	-0.18
0.405	0.9195	-0.3352	-0.19
0.410	0.9178	-0.3388	-0.19
0.415	0.9160	-0.3425	-0.19
0.420	0.9141	-0.3461	-0.20
0.425	0.9123	-0.3498	-0.20
0.430	0.9106	-0.3532	-0.20
0.435	0.9087	-0.3568	-0.21
0.440	0.9069	-0.3603	-0.21
0.445	0.9049	-0.3638	-0.22
0.450	0.9031	-0.3674	-0.22
0.455	0.9011	-0.3708	-0.23
0.460	0.8992	-0.3744	-0.23
0.465	0.8971	-0.3777	-0.23
0.470	0.8952	-0.3812	-0.24
0.475	0.8931	-0.3846	-0.24
0.480	0.8913	-0.3880	-0.25
0.485	0.8893	-0.3913	-0.25
0.490	0.8873	-0.3947	-0.25
0.495	0.8854	-0.3980	-0.26
0.500	0.8834	-0.4014	-0.26
0.505	0.8813	-0.4047	-0.27
0.510	0.8792	-0.4080	-0.27
0.515	0.8773	-0.4113	-0.27
0.520	0.8755	-0.4145	-0.28
0.525	0.8735	-0.4177	-0.28
0.530	0.8715	-0.4208	-0.28
0.535	0.8696	-0.4241	-0.29
0.540	0.8674	-0.4273	-0.29
0.545	0.8653	-0.4305	-0.30
0.550	0.8632	-0.4337	-0.30
0.555	0.8611	-0.4369	-0.30
0.560	0.8590	-0.4400	-0.31
0.565	0.8570	-0.4430	-0.31
0.570	0.8549	-0.4461	-0.32
0.575	0.8527	-0.4494	-0.32

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8506	-0.4524	-0.32
0.585	0.8485	-0.4555	-0.33
0.590	0.8463	-0.4585	-0.33
0.595	0.8443	-0.4615	-0.33
0.600	0.8420	-0.4645	-0.34
0.605	0.8398	-0.4675	-0.34
0.610	0.8376	-0.4703	-0.35
0.615	0.8354	-0.4733	-0.35
0.620	0.8332	-0.4763	-0.36
0.625	0.8310	-0.4791	-0.36
0.630	0.8288	-0.4820	-0.37
0.635	0.8265	-0.4850	-0.37
0.640	0.8243	-0.4878	-0.37
0.645	0.8221	-0.4907	-0.38
0.650	0.8197	-0.4935	-0.38
0.655	0.8174	-0.4963	-0.39
0.660	0.8151	-0.4991	-0.39
0.665	0.8128	-0.5019	-0.40
0.670	0.8104	-0.5045	-0.40
0.675	0.8081	-0.5073	-0.41
0.680	0.8057	-0.5099	-0.41
0.685	0.8033	-0.5125	-0.42
0.690	0.8008	-0.5151	-0.43
0.695	0.7983	-0.5175	-0.43
0.700	0.7957	-0.5202	-0.44
0.705	0.7934	-0.5227	-0.44
0.710	0.7909	-0.5253	-0.45
0.715	0.7885	-0.5280	-0.46
0.720	0.7860	-0.5304	-0.46
0.725	0.7834	-0.5329	-0.47
0.730	0.7809	-0.5354	-0.47
0.735	0.7782	-0.5378	-0.48
0.740	0.7758	-0.5403	-0.49
0.745	0.7732	-0.5426	-0.50
0.750	0.7707	-0.5450	-0.50
0.755	0.7681	-0.5473	-0.51
0.760	0.7655	-0.5495	-0.52
0.765	0.7629	-0.5519	-0.52
0.770	0.7601	-0.5542	-0.53
0.775	0.7574	-0.5564	-0.54
0.780	0.7549	-0.5586	-0.55
0.785	0.7522	-0.5610	-0.55
0.790	0.7495	-0.5631	-0.56
0.795	0.7469	-0.5653	-0.57
0.800	0.7442	-0.5672	-0.58
0.805	0.7414	-0.5692	-0.59

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.7387	-0.5714	-0.59
0.815	0.7362	-0.5733	-0.60
0.820	0.7336	-0.5754	-0.61
0.825	0.7310	-0.5772	-0.62
0.830	0.7285	-0.5794	-0.62
0.835	0.7259	-0.5813	-0.63
0.840	0.7235	-0.5833	-0.64
0.845	0.7210	-0.5855	-0.64
0.850	0.7185	-0.5875	-0.65
0.855	0.7158	-0.5895	-0.66
0.860	0.7133	-0.5916	-0.66
0.865	0.7107	-0.5936	-0.67
0.870	0.7080	-0.5957	-0.67
0.875	0.7055	-0.5977	-0.68
0.880	0.7029	-0.5998	-0.69
0.885	0.7004	-0.6019	-0.69
0.890	0.6976	-0.6039	-0.70
0.895	0.6951	-0.6060	-0.70
0.900	0.6924	-0.6081	-0.71
0.905	0.6897	-0.6101	-0.72
0.910	0.6870	-0.6121	-0.72
0.915	0.6844	-0.6142	-0.73
0.920	0.6816	-0.6162	-0.74
0.925	0.6788	-0.6182	-0.74
0.930	0.6760	-0.6202	-0.75
0.935	0.6734	-0.6223	-0.75
0.940	0.6705	-0.6243	-0.76
0.945	0.6678	-0.6262	-0.77
0.950	0.6651	-0.6283	-0.77
0.955	0.6621	-0.6303	-0.78
0.960	0.6594	-0.6322	-0.79
0.965	0.6566	-0.6342	-0.79
0.970	0.6538	-0.6362	-0.80
0.975	0.6510	-0.6382	-0.80
0.980	0.6483	-0.6401	-0.81
0.985	0.6454	-0.6419	-0.82
0.990	0.6426	-0.6438	-0.82
0.995	0.6398	-0.6457	-0.83
1.000	0.6370	-0.6474	-0.84
1.005	0.6340	-0.6493	-0.84
1.010	0.6312	-0.6511	-0.85
1.015	0.6284	-0.6528	-0.86
1.020	0.6254	-0.6546	-0.86
1.025	0.6226	-0.6564	-0.87
1.030	0.6197	-0.6582	-0.88
1.035	0.6169	-0.6600	-0.88

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.6140	-0.6617	-0.89
1.045	0.6113	-0.6633	-0.90
1.050	0.6085	-0.6650	-0.90
1.055	0.6058	-0.6665	-0.91
1.060	0.6031	-0.6682	-0.91
1.065	0.6003	-0.6699	-0.92
1.070	0.5974	-0.6716	-0.93
1.075	0.5947	-0.6731	-0.93
1.080	0.5918	-0.6747	-0.94
1.085	0.5892	-0.6763	-0.95
1.090	0.5864	-0.6777	-0.95
1.095	0.5836	-0.6791	-0.96
1.100	0.5810	-0.6807	-0.96
1.105	0.5781	-0.6822	-0.97
1.110	0.5753	-0.6836	-0.98
1.115	0.5727	-0.6851	-0.98
1.120	0.5699	-0.6863	-0.99
1.125	0.5673	-0.6878	-1.00
1.130	0.5647	-0.6891	-1.00
1.135	0.5620	-0.6904	-1.01
1.140	0.5593	-0.6919	-1.01
1.145	0.5568	-0.6932	-1.02
1.150	0.5542	-0.6945	-1.03
1.155	0.5516	-0.6959	-1.03
1.160	0.5490	-0.6970	-1.04
1.165	0.5465	-0.6984	-1.04
1.170	0.5439	-0.6995	-1.05
1.175	0.5414	-0.7009	-1.05
1.180	0.5390	-0.7020	-1.06
1.185	0.5365	-0.7031	-1.07
1.190	0.5342	-0.7044	-1.07
1.195	0.5318	-0.7054	-1.08
1.200	0.5294	-0.7065	-1.08
1.205	0.5270	-0.7074	-1.09
1.210	0.5246	-0.7085	-1.09
1.215	0.5222	-0.7096	-1.10
1.220	0.5197	-0.7106	-1.11
1.225	0.5173	-0.7116	-1.11
1.230	0.5149	-0.7126	-1.12
1.235	0.5126	-0.7136	-1.12
1.240	0.5102	-0.7145	-1.13
1.245	0.5078	-0.7156	-1.14
1.250	0.5055	-0.7165	-1.14
1.255	0.5032	-0.7174	-1.15
1.260	0.5008	-0.7182	-1.15
1.265	0.4986	-0.7192	-1.16

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.4963	-0.7202	-1.16
1.275	0.4939	-0.7211	-1.17
1.280	0.4916	-0.7219	-1.18
1.285	0.4891	-0.7229	-1.18
1.290	0.4868	-0.7236	-1.19
1.295	0.4843	-0.7246	-1.19
1.300	0.4819	-0.7255	-1.20
1.305	0.4797	-0.7263	-1.21
1.310	0.4775	-0.7271	-1.21
1.315	0.4751	-0.7279	-1.22
1.320	0.4728	-0.7287	-1.22
1.325	0.4706	-0.7293	-1.23
1.330	0.4684	-0.7302	-1.23
1.335	0.4662	-0.7309	-1.24
1.340	0.4638	-0.7317	-1.25
1.345	0.4615	-0.7323	-1.25
1.350	0.4591	-0.7327	-1.26
1.355	0.4569	-0.7336	-1.27
1.360	0.4545	-0.7344	-1.27
1.365	0.4521	-0.7350	-1.28
1.370	0.4497	-0.7357	-1.29
1.375	0.4472	-0.7363	-1.30
1.380	0.4447	-0.7367	-1.30
1.385	0.4422	-0.7375	-1.31
1.390	0.4397	-0.7382	-1.32
1.395	0.4373	-0.7386	-1.33
1.400	0.4348	-0.7393	-1.33
1.405	0.4322	-0.7399	-1.34
1.410	0.4295	-0.7403	-1.35
1.415	0.4270	-0.7410	-1.36
1.420	0.4244	-0.7417	-1.37
1.425	0.4217	-0.7423	-1.37
1.430	0.4192	-0.7429	-1.38
1.435	0.4166	-0.7436	-1.39
1.440	0.4138	-0.7441	-1.40
1.445	0.4111	-0.7447	-1.41
1.450	0.4085	-0.7453	-1.41
1.455	0.4058	-0.7458	-1.42
1.460	0.4030	-0.7463	-1.43
1.465	0.4004	-0.7468	-1.44
1.470	0.3977	-0.7473	-1.45
1.475	0.3948	-0.7479	-1.46
1.480	0.3921	-0.7483	-1.47
1.485	0.3892	-0.7488	-1.47
1.490	0.3863	-0.7494	-1.48
1.495	0.3836	-0.7499	-1.49

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.500	0.3808	-0.7503	-1.50
1.505	0.3779	-0.7505	-1.51
1.510	0.3751	-0.7508	-1.52
1.515	0.3722	-0.7513	-1.53
1.520	0.3693	-0.7519	-1.54
1.525	0.3663	-0.7523	-1.55
1.530	0.3634	-0.7526	-1.56
1.535	0.3604	-0.7530	-1.57
1.540	0.3575	-0.7533	-1.58
1.545	0.3546	-0.7536	-1.59
1.550	0.3516	-0.7539	-1.60
1.555	0.3486	-0.7543	-1.61
1.560	0.3457	-0.7546	-1.62
1.565	0.3426	-0.7549	-1.63
1.570	0.3396	-0.7552	-1.64
1.575	0.3365	-0.7556	-1.65
1.580	0.3337	-0.7558	-1.66
1.585	0.3306	-0.7562	-1.67
1.590	0.3278	-0.7567	-1.67
1.595	0.3247	-0.7571	-1.68
1.600	0.3218	-0.7573	-1.69
1.605	0.3190	-0.7578	-1.70
1.610	0.3162	-0.7580	-1.71
1.615	0.3131	-0.7581	-1.72
1.620	0.3101	-0.7583	-1.73
1.625	0.3070	-0.7586	-1.74
1.630	0.3042	-0.7589	-1.75
1.635	0.3013	-0.7591	-1.76
1.640	0.2982	-0.7592	-1.77
1.645	0.2953	-0.7593	-1.78
1.650	0.2922	-0.7596	-1.79
1.655	0.2892	-0.7599	-1.80
1.660	0.2865	-0.7602	-1.81
1.665	0.2837	-0.7602	-1.81
1.670	0.2808	-0.7603	-1.82
1.675	0.2780	-0.7604	-1.83
1.680	0.2752	-0.7606	-1.84
1.685	0.2724	-0.7608	-1.85
1.690	0.2698	-0.7609	-1.86
1.695	0.2672	-0.7610	-1.87
1.700	0.2644	-0.7609	-1.88
1.705	0.2616	-0.7610	-1.89
1.710	0.2590	-0.7612	-1.89
1.715	0.2562	-0.7612	-1.90
1.720	0.2534	-0.7613	-1.91
1.725	0.2508	-0.7613	-1.92

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.2481	-0.7614	-1.93
1.735	0.2455	-0.7614	-1.94
1.740	0.2431	-0.7613	-1.95
1.745	0.2407	-0.7616	-1.95
1.750	0.2381	-0.7617	-1.96
1.755	0.2359	-0.7617	-1.97
1.760	0.2337	-0.7618	-1.97
1.765	0.2315	-0.7620	-1.98
1.770	0.2291	-0.7620	-1.98
1.775	0.2271	-0.7620	-1.99
1.780	0.2251	-0.7618	-2.00
1.785	0.2229	-0.7617	-2.01
1.790	0.2207	-0.7615	-2.02
1.795	0.2185	-0.7615	-2.02
1.800	0.2163	-0.7615	-2.03
1.805	0.2144	-0.7616	-2.03
1.810	0.2123	-0.7617	-2.04
1.815	0.2103	-0.7618	-2.04
1.820	0.2085	-0.7616	-2.05
1.825	0.2067	-0.7616	-2.06
1.830	0.2050	-0.7614	-2.06
1.835	0.2029	-0.7615	-2.07
1.840	0.2012	-0.7615	-2.07
1.845	0.1995	-0.7612	-2.08
1.850	0.1978	-0.7611	-2.09
1.855	0.1959	-0.7609	-2.09
1.860	0.1941	-0.7609	-2.10
1.865	0.1924	-0.7607	-2.11
1.870	0.1907	-0.7608	-2.11
1.875	0.1888	-0.7609	-2.11
1.880	0.1873	-0.7612	-2.11
1.885	0.1858	-0.7614	-2.12
1.890	0.1845	-0.7613	-2.12
1.895	0.1830	-0.7611	-2.13
1.900	0.1815	-0.7609	-2.13
1.905	0.1799	-0.7612	-2.13
1.910	0.1784	-0.7611	-2.14
1.915	0.1768	-0.7614	-2.14
1.920	0.1754	-0.7614	-2.14
1.925	0.1741	-0.7615	-2.15
1.930	0.1725	-0.7615	-2.15
1.935	0.1712	-0.7616	-2.15
1.940	0.1699	-0.7616	-2.15
1.945	0.1685	-0.7616	-2.16
1.950	0.1673	-0.7616	-2.16
1.955	0.1658	-0.7617	-2.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.1643	-0.7617	-2.17
1.965	0.1631	-0.7616	-2.17
1.970	0.1618	-0.7616	-2.17
1.975	0.1607	-0.7616	-2.18
1.980	0.1592	-0.7616	-2.18
1.985	0.1579	-0.7614	-2.18
1.990	0.1564	-0.7616	-2.19
1.995	0.1551	-0.7614	-2.19
2.000	0.1537	-0.7616	-2.19
2.005	0.1523	-0.7615	-2.20
2.010	0.1509	-0.7614	-2.20
2.015	0.1495	-0.7614	-2.20
2.020	0.1481	-0.7613	-2.21
2.025	0.1465	-0.7614	-2.21
2.030	0.1450	-0.7615	-2.21
2.035	0.1436	-0.7613	-2.22
2.040	0.1421	-0.7611	-2.22
2.045	0.1405	-0.7612	-2.22
2.050	0.1389	-0.7610	-2.23
2.055	0.1374	-0.7609	-2.23
2.060	0.1357	-0.7609	-2.24
2.065	0.1340	-0.7607	-2.24
2.070	0.1324	-0.7607	-2.25
2.075	0.1307	-0.7605	-2.25
2.080	0.1288	-0.7605	-2.26
2.085	0.1270	-0.7603	-2.26
2.090	0.1253	-0.7600	-2.27
2.095	0.1233	-0.7597	-2.27
2.100	0.1213	-0.7598	-2.28
2.105	0.1193	-0.7596	-2.28
2.110	0.1173	-0.7594	-2.29
2.115	0.1152	-0.7594	-2.29
2.120	0.1131	-0.7595	-2.29
2.125	0.1111	-0.7592	-2.30
2.130	0.1090	-0.7590	-2.31
2.135	0.1069	-0.7587	-2.31
2.140	0.1047	-0.7588	-2.32
2.145	0.1025	-0.7586	-2.32
2.150	0.1002	-0.7583	-2.33
2.155	0.0981	-0.7581	-2.33
2.160	0.0960	-0.7579	-2.34
2.165	0.0936	-0.7576	-2.35
2.170	0.0914	-0.7575	-2.35
2.175	0.0890	-0.7572	-2.36
2.180	0.0868	-0.7569	-2.36
2.185	0.0845	-0.7566	-2.37

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.0821	-0.7562	-2.38
2.195	0.0797	-0.7560	-2.38
2.200	0.0773	-0.7559	-2.39
2.205	0.0749	-0.7553	-2.39
2.210	0.0725	-0.7549	-2.40
2.215	0.0701	-0.7547	-2.41
2.220	0.0677	-0.7545	-2.41
2.225	0.0652	-0.7541	-2.42
2.230	0.0628	-0.7534	-2.43
2.235	0.0603	-0.7534	-2.43
2.240	0.0578	-0.7531	-2.44
2.245	0.0555	-0.7528	-2.44
2.250	0.0531	-0.7523	-2.45
2.255	0.0507	-0.7521	-2.45
2.260	0.0484	-0.7519	-2.46
2.265	0.0460	-0.7514	-2.47
2.270	0.0437	-0.7510	-2.47
2.275	0.0412	-0.7507	-2.48
2.280	0.0389	-0.7503	-2.48
2.285	0.0365	-0.7501	-2.49
2.290	0.0343	-0.7494	-2.50
2.295	0.0320	-0.7491	-2.50
2.300	0.0298	-0.7486	-2.51
2.305	0.0273	-0.7483	-2.51
2.310	0.0250	-0.7479	-2.52
2.315	0.0229	-0.7474	-2.52
2.320	0.0206	-0.7470	-2.53
2.325	0.0184	-0.7466	-2.54
2.330	0.0160	-0.7462	-2.54
2.335	0.0141	-0.7457	-2.55
2.340	0.0118	-0.7454	-2.55
2.345	0.0097	-0.7448	-2.56
2.350	0.0077	-0.7443	-2.56
2.355	0.0057	-0.7438	-2.57
2.360	0.0039	-0.7435	-2.57
2.365	0.0018	-0.7429	-2.58
2.370	-0.0002	-0.7422	-2.59
2.375	-0.0018	-0.7418	-2.59
2.380	-0.0038	-0.7414	-2.60
2.385	-0.0056	-0.7410	-2.60
2.390	-0.0071	-0.7406	-2.61
2.395	-0.0090	-0.7402	-2.61
2.400	-0.0107	-0.7398	-2.62
2.405	-0.0123	-0.7394	-2.62
2.410	-0.0139	-0.7389	-2.63
2.415	-0.0155	-0.7385	-2.63

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.420	-0.0171	-0.7381	-2.64
2.425	-0.0185	-0.7377	-2.64
2.430	-0.0198	-0.7372	-2.65
2.435	-0.0213	-0.7367	-2.65
2.440	-0.0226	-0.7363	-2.66
2.445	-0.0239	-0.7358	-2.66
2.450	-0.0251	-0.7354	-2.66
2.455	-0.0262	-0.7349	-2.67
2.460	-0.0273	-0.7345	-2.67
2.465	-0.0284	-0.7340	-2.68
2.470	-0.0297	-0.7335	-2.69
2.475	-0.0307	-0.7332	-2.69
2.480	-0.0319	-0.7328	-2.69
2.485	-0.0330	-0.7325	-2.70
2.490	-0.0339	-0.7320	-2.70
2.495	-0.0351	-0.7316	-2.70
2.500	-0.0362	-0.7313	-2.71
2.505	-0.0372	-0.7311	-2.71
2.510	-0.0379	-0.7309	-2.71
2.515	-0.0392	-0.7306	-2.71
2.520	-0.0399	-0.7303	-2.72
2.525	-0.0411	-0.7300	-2.72
2.530	-0.0420	-0.7298	-2.72
2.535	-0.0431	-0.7296	-2.72
2.540	-0.0440	-0.7295	-2.72
2.545	-0.0449	-0.7292	-2.73
2.550	-0.0458	-0.7289	-2.73
2.555	-0.0465	-0.7285	-2.73
2.560	-0.0472	-0.7280	-2.74
2.565	-0.0476	-0.7276	-2.74
2.570	-0.0480	-0.7270	-2.75
2.575	-0.0484	-0.7265	-2.76
2.580	-0.0486	-0.7261	-2.76
2.585	-0.0489	-0.7258	-2.76
2.590	-0.0492	-0.7256	-2.77
2.595	-0.0494	-0.7255	-2.77
2.600	-0.0499	-0.7254	-2.77
2.605	-0.0502	-0.7253	-2.77
2.610	-0.0505	-0.7252	-2.77
2.615	-0.0510	-0.7249	-2.77
2.620	-0.0514	-0.7247	-2.78
2.625	-0.0517	-0.7246	-2.78
2.630	-0.0520	-0.7243	-2.78
2.635	-0.0525	-0.7240	-2.78
2.640	-0.0528	-0.7238	-2.78
2.645	-0.0532	-0.7235	-2.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	-0.0536	-0.7233	-2.79
2.655	-0.0542	-0.7230	-2.79
2.660	-0.0545	-0.7228	-2.79
2.665	-0.0549	-0.7225	-2.80
2.670	-0.0555	-0.7223	-2.80
2.675	-0.0563	-0.7222	-2.80
2.680	-0.0567	-0.7220	-2.80
2.685	-0.0573	-0.7220	-2.80
2.690	-0.0580	-0.7220	-2.80
2.695	-0.0583	-0.7217	-2.80
2.700	-0.0588	-0.7215	-2.81
2.705	-0.0592	-0.7211	-2.81
2.710	-0.0599	-0.7212	-2.81
2.715	-0.0602	-0.7206	-2.82
2.720	-0.0606	-0.7202	-2.82
2.725	-0.0612	-0.7196	-2.83
2.730	-0.0616	-0.7189	-2.83
2.735	-0.0622	-0.7184	-2.84
2.740	-0.0628	-0.7176	-2.85
2.745	-0.0637	-0.7170	-2.86
2.750	-0.0645	-0.7163	-2.86
2.755	-0.0654	-0.7156	-2.87
2.760	-0.0663	-0.7149	-2.88
2.765	-0.0673	-0.7142	-2.89
2.770	-0.0681	-0.7134	-2.89
2.775	-0.0692	-0.7128	-2.90
2.780	-0.0702	-0.7120	-2.91
2.785	-0.0714	-0.7116	-2.91
2.790	-0.0724	-0.7109	-2.92
2.795	-0.0736	-0.7104	-2.92
2.800	-0.0746	-0.7097	-2.93
2.805	-0.0758	-0.7091	-2.94
2.810	-0.0772	-0.7086	-2.94
2.815	-0.0783	-0.7082	-2.94
2.820	-0.0795	-0.7077	-2.95
2.825	-0.0806	-0.7070	-2.96
2.830	-0.0815	-0.7066	-2.96
2.835	-0.0828	-0.7061	-2.96
2.840	-0.0840	-0.7056	-2.97
2.845	-0.0851	-0.7051	-2.97
2.850	-0.0861	-0.7045	-2.98
2.855	-0.0871	-0.7040	-2.98
2.860	-0.0882	-0.7035	-2.99
2.865	-0.0892	-0.7029	-2.99
2.870	-0.0901	-0.7024	-3.00
2.875	-0.0909	-0.7017	-3.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	-0.0917	-0.7008	-3.01
2.885	-0.0923	-0.6997	-3.03
2.890	-0.0937	-0.6988	-3.04
2.895	-0.0947	-0.6977	-3.05
2.900	-0.0962	-0.6968	-3.06
2.905	-0.0973	-0.6959	-3.07
2.910	-0.0987	-0.6949	-3.07
2.915	-0.1004	-0.6943	-3.08
2.920	-0.1018	-0.6936	-3.09
2.925	-0.1032	-0.6931	-3.09
2.930	-0.1048	-0.6925	-3.09
2.935	-0.1061	-0.6922	-3.09
2.940	-0.1072	-0.6919	-3.10
2.945	-0.1085	-0.6919	-3.09
2.950	-0.1097	-0.6916	-3.09
2.955	-0.1107	-0.6915	-3.09
2.960	-0.1115	-0.6914	-3.09
2.965	-0.1123	-0.6915	-3.09
2.970	-0.1130	-0.6914	-3.09
2.975	-0.1137	-0.6914	-3.09
2.980	-0.1144	-0.6915	-3.09
2.985	-0.1146	-0.6913	-3.09
2.990	-0.1150	-0.6913	-3.09
2.995	-0.1154	-0.6911	-3.09
3.000	-0.1157	-0.6906	-3.10
3.005	-0.1187	-0.6896	-3.10
3.010	-0.1193	-0.6898	-3.10
3.015	-0.1193	-0.6897	-3.10
3.020	-0.1197	-0.6899	-3.10
3.025	-0.1196	-0.6899	-3.10
3.030	-0.1195	-0.6895	-3.10
3.035	-0.1193	-0.6888	-3.11
3.040	-0.1194	-0.6883	-3.12
3.045	-0.1192	-0.6876	-3.13
3.050	-0.1190	-0.6869	-3.13
3.055	-0.1193	-0.6860	-3.14
3.060	-0.1196	-0.6852	-3.15
3.065	-0.1200	-0.6840	-3.17
3.070	-0.1205	-0.6831	-3.18
3.075	-0.1211	-0.6823	-3.19
3.080	-0.1215	-0.6813	-3.20
3.085	-0.1224	-0.6802	-3.21
3.090	-0.1236	-0.6794	-3.22
3.095	-0.1247	-0.6790	-3.22
3.100	-0.1259	-0.6788	-3.22
3.105	-0.1269	-0.6784	-3.22

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.110	-0.1280	-0.6783	-3.22
3.115	-0.1292	-0.6781	-3.22
3.120	-0.1303	-0.6783	-3.21
3.125	-0.1312	-0.6787	-3.21
3.130	-0.1322	-0.6789	-3.20
3.135	-0.1329	-0.6796	-3.19
3.140	-0.1335	-0.6800	-3.19
3.145	-0.1341	-0.6806	-3.18
3.150	-0.1345	-0.6812	-3.17
3.155	-0.1348	-0.6815	-3.16
3.160	-0.1348	-0.6819	-3.16
3.165	-0.1349	-0.6824	-3.15
3.170	-0.1349	-0.6825	-3.15
3.175	-0.1349	-0.6828	-3.15
3.180	-0.1343	-0.6828	-3.15
3.185	-0.1343	-0.6825	-3.15
3.190	-0.1340	-0.6821	-3.16
3.195	-0.1340	-0.6813	-3.17
3.200	-0.1336	-0.6806	-3.18
3.205	-0.1340	-0.6796	-3.19
3.210	-0.1343	-0.6787	-3.20
3.215	-0.1346	-0.6776	-3.21
3.220	-0.1352	-0.6765	-3.22
3.225	-0.1363	-0.6760	-3.23
3.230	-0.1374	-0.6759	-3.23
3.235	-0.1386	-0.6757	-3.23
3.240	-0.1394	-0.6755	-3.23
3.245	-0.1402	-0.6759	-3.22
3.250	-0.1413	-0.6758	-3.22
3.255	-0.1419	-0.6762	-3.21
3.260	-0.1429	-0.6763	-3.21
3.265	-0.1434	-0.6763	-3.21
3.270	-0.1443	-0.6765	-3.20
3.275	-0.1448	-0.6768	-3.20
3.280	-0.1454	-0.6769	-3.19
3.285	-0.1462	-0.6769	-3.19
3.290	-0.1472	-0.6768	-3.19
3.295	-0.1475	-0.6770	-3.19
3.300	-0.1480	-0.6769	-3.19
3.305	-0.1484	-0.6767	-3.19
3.310	-0.1492	-0.6768	-3.18
3.315	-0.1495	-0.6764	-3.19
3.320	-0.1498	-0.6763	-3.19
3.325	-0.1503	-0.6762	-3.19
3.330	-0.1505	-0.6756	-3.20
3.335	-0.1509	-0.6752	-3.20

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	-0.1514	-0.6748	-3.20
3.345	-0.1516	-0.6743	-3.21
3.350	-0.1522	-0.6735	-3.22
3.355	-0.1526	-0.6726	-3.23
3.360	-0.1528	-0.6717	-3.24
3.365	-0.1536	-0.6710	-3.24
3.370	-0.1541	-0.6699	-3.26
3.375	-0.1545	-0.6685	-3.27
3.380	-0.1552	-0.6675	-3.28
3.385	-0.1561	-0.6664	-3.29
3.390	-0.1568	-0.6650	-3.31
3.395	-0.1576	-0.6637	-3.32
3.400	-0.1586	-0.6624	-3.34
3.405	-0.1593	-0.6608	-3.35
3.410	-0.1607	-0.6597	-3.36
3.415	-0.1617	-0.6581	-3.38
3.420	-0.1631	-0.6572	-3.39
3.425	-0.1646	-0.6559	-3.40
3.430	-0.1657	-0.6544	-3.41
3.435	-0.1671	-0.6535	-3.42
3.440	-0.1685	-0.6522	-3.43
3.445	-0.1703	-0.6509	-3.44
3.450	-0.1718	-0.6499	-3.45
3.455	-0.1737	-0.6492	-3.45
3.460	-0.1748	-0.6483	-3.46
3.465	-0.1765	-0.6480	-3.46
3.470	-0.1780	-0.6474	-3.46
3.475	-0.1793	-0.6468	-3.46
3.480	-0.1807	-0.6465	-3.46
3.485	-0.1820	-0.6461	-3.46
3.490	-0.1834	-0.6464	-3.45
3.495	-0.1841	-0.6463	-3.45
3.500	-0.1848	-0.6462	-3.45
3.505	-0.1854	-0.6464	-3.45
3.510	-0.1856	-0.6458	-3.45
3.515	-0.1859	-0.6454	-3.46
3.520	-0.1858	-0.6447	-3.47
3.525	-0.1860	-0.6442	-3.47
3.530	-0.1859	-0.6433	-3.48
3.535	-0.1850	-0.6416	-3.51
3.540	-0.1844	-0.6398	-3.53
3.545	-0.1846	-0.6373	-3.56
3.550	-0.1847	-0.6346	-3.60
3.555	-0.1855	-0.6321	-3.63
3.560	-0.1870	-0.6290	-3.66
3.565	-0.1883	-0.6266	-3.69

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.570	-0.1909	-0.6240	-3.71
3.575	-0.1939	-0.6218	-3.72
3.580	-0.1971	-0.6196	-3.74
3.585	-0.2002	-0.6187	-3.74
3.590	-0.2040	-0.6196	-3.71
3.595	-0.2067	-0.6203	-3.69
3.600	-0.2085	-0.6221	-3.66
3.605	-0.2099	-0.6238	-3.63
3.610	-0.2099	-0.6258	-3.61
3.615	-0.2096	-0.6268	-3.60
3.620	-0.2082	-0.6272	-3.60
3.625	-0.2073	-0.6274	-3.60
3.630	-0.2060	-0.6274	-3.60
3.635	-0.2045	-0.6265	-3.62
3.640	-0.2039	-0.6251	-3.64
3.645	-0.2037	-0.6233	-3.67
3.650	-0.2036	-0.6217	-3.69
3.655	-0.2036	-0.6202	-3.70
3.660	-0.2041	-0.6187	-3.72
3.665	-0.2044	-0.6176	-3.73
3.670	-0.2050	-0.6158	-3.75
3.675	-0.2052	-0.6141	-3.78
3.680	-0.2057	-0.6124	-3.80
3.685	-0.2065	-0.6107	-3.81
3.690	-0.2074	-0.6090	-3.83
3.695	-0.2090	-0.6071	-3.85
3.700	-0.2107	-0.6051	-3.87
3.705	-0.2125	-0.6035	-3.88
3.710	-0.2152	-0.6015	-3.89
3.715	-0.2183	-0.6004	-3.89
3.720	-0.2219	-0.5994	-3.89
3.725	-0.2255	-0.5996	-3.87
3.730	-0.2300	-0.6002	-3.84
3.735	-0.2331	-0.6022	-3.80
3.740	-0.2359	-0.6048	-3.75
3.745	-0.2373	-0.6076	-3.71
3.750	-0.2376	-0.6111	-3.67
3.755	-0.2368	-0.6139	-3.64
3.760	-0.2349	-0.6164	-3.61
3.765	-0.2319	-0.6178	-3.61
3.770	-0.2287	-0.6175	-3.63
3.775	-0.2258	-0.6164	-3.66
3.780	-0.2232	-0.6135	-3.70
3.785	-0.2230	-0.6095	-3.75
3.790	-0.2237	-0.6060	-3.80
3.795	-0.2258	-0.6035	-3.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	-0.2280	-0.6015	-3.83
3.805	-0.2306	-0.5997	-3.84
3.810	-0.2329	-0.5991	-3.84
3.815	-0.2355	-0.5984	-3.83
3.820	-0.2375	-0.5982	-3.83
3.825	-0.2397	-0.5982	-3.82
3.830	-0.2416	-0.5986	-3.80
3.835	-0.2424	-0.6016	-3.76
3.840	-0.2435	-0.6013	-3.76
3.845	-0.2445	-0.6012	-3.76
3.850	-0.2453	-0.6008	-3.76
3.855	-0.2464	-0.6004	-3.76
3.860	-0.2474	-0.6001	-3.75
3.865	-0.2483	-0.5995	-3.76
3.870	-0.2493	-0.5992	-3.76
3.875	-0.2505	-0.5987	-3.75
3.880	-0.2518	-0.5981	-3.76
3.885	-0.2530	-0.5976	-3.76
3.890	-0.2543	-0.5971	-3.76
3.895	-0.2556	-0.5964	-3.76
3.900	-0.2568	-0.5960	-3.76
3.905	-0.2584	-0.5953	-3.76
3.910	-0.2596	-0.5948	-3.76
3.915	-0.2612	-0.5942	-3.75
3.920	-0.2627	-0.5935	-3.76
3.925	-0.2642	-0.5929	-3.75
3.930	-0.2654	-0.5922	-3.76
3.935	-0.2670	-0.5913	-3.76
3.940	-0.2684	-0.5907	-3.76
3.945	-0.2699	-0.5901	-3.76
3.950	-0.2713	-0.5891	-3.76
3.955	-0.2726	-0.5884	-3.76
3.960	-0.2742	-0.5874	-3.77
3.965	-0.2754	-0.5864	-3.77
3.970	-0.2770	-0.5855	-3.77
3.975	-0.2782	-0.5845	-3.78
3.980	-0.2800	-0.5837	-3.78
3.985	-0.2813	-0.5825	-3.78
3.990	-0.2828	-0.5816	-3.79
3.995	-0.2844	-0.5806	-3.79
4.000	-0.2859	-0.5796	-3.79
4.005	-0.2874	-0.5786	-3.80
4.010	-0.2887	-0.5776	-3.80
4.015	-0.2900	-0.5763	-3.81
4.020	-0.2915	-0.5754	-3.81
4.025	-0.2927	-0.5745	-3.81

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	-0.2942	-0.5733	-3.82
4.035	-0.2957	-0.5721	-3.82
4.040	-0.2972	-0.5710	-3.83
4.045	-0.2986	-0.5699	-3.83
4.050	-0.2998	-0.5689	-3.84
4.055	-0.3010	-0.5675	-3.84
4.060	-0.3023	-0.5664	-3.85
4.065	-0.3036	-0.5653	-3.85
4.070	-0.3048	-0.5644	-3.86
4.075	-0.3060	-0.5634	-3.86
4.080	-0.3071	-0.5625	-3.87
4.085	-0.3085	-0.5615	-3.87
4.090	-0.3096	-0.5605	-3.87
4.095	-0.3108	-0.5598	-3.87
4.100	-0.3118	-0.5586	-3.88
4.105	-0.3131	-0.5578	-3.88
4.110	-0.3142	-0.5573	-3.88
4.115	-0.3155	-0.5563	-3.88
4.120	-0.3169	-0.5555	-3.88
4.125	-0.3182	-0.5549	-3.88
4.130	-0.3195	-0.5539	-3.88
4.135	-0.3206	-0.5528	-3.89
4.140	-0.3220	-0.5517	-3.89
4.145	-0.3232	-0.5509	-3.89
4.150	-0.3242	-0.5494	-3.90
4.155	-0.3248	-0.5483	-3.91
4.160	-0.3255	-0.5473	-3.92
4.165	-0.3261	-0.5465	-3.93
4.170	-0.3262	-0.5455	-3.94
4.175	-0.3263	-0.5450	-3.94
4.180	-0.3265	-0.5444	-3.95
4.185	-0.3268	-0.5439	-3.95
4.190	-0.3271	-0.5436	-3.95
4.195	-0.3272	-0.5433	-3.96
4.200	-0.3273	-0.5431	-3.96
4.205	-0.3276	-0.5432	-3.95
4.210	-0.3276	-0.5429	-3.96
4.215	-0.3279	-0.5427	-3.96
4.220	-0.3279	-0.5429	-3.95
4.225	-0.3280	-0.5426	-3.96
4.230	-0.3282	-0.5426	-3.96
4.235	-0.3281	-0.5428	-3.95
4.240	-0.3283	-0.5429	-3.95
4.245	-0.3283	-0.5434	-3.95
4.250	-0.3284	-0.5434	-3.95
4.255	-0.3281	-0.5435	-3.95

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.260	-0.3279	-0.5437	-3.95
4.265	-0.3279	-0.5439	-3.94
4.270	-0.3279	-0.5442	-3.94
4.275	-0.3277	-0.5445	-3.94
4.280	-0.3275	-0.5449	-3.93
4.285	-0.3274	-0.5450	-3.93
4.290	-0.3273	-0.5453	-3.93
4.295	-0.3269	-0.5454	-3.93
4.300	-0.3271	-0.5460	-3.92
4.305	-0.3271	-0.5464	-3.92
4.310	-0.3270	-0.5465	-3.92
4.315	-0.3269	-0.5469	-3.92
4.320	-0.3268	-0.5472	-3.91
4.325	-0.3268	-0.5476	-3.91
4.330	-0.3266	-0.5480	-3.90
4.335	-0.3266	-0.5486	-3.90
4.340	-0.3266	-0.5490	-3.89
4.345	-0.3265	-0.5498	-3.88
4.350	-0.3264	-0.5503	-3.88
4.355	-0.3262	-0.5509	-3.87
4.360	-0.3259	-0.5516	-3.87
4.365	-0.3255	-0.5524	-3.86
4.370	-0.3253	-0.5530	-3.86
4.375	-0.3248	-0.5536	-3.85
4.380	-0.3242	-0.5544	-3.85
4.385	-0.3239	-0.5550	-3.84
4.390	-0.3232	-0.5556	-3.84
4.395	-0.3227	-0.5562	-3.84
4.400	-0.3222	-0.5566	-3.83
4.405	-0.3215	-0.5571	-3.83
4.410	-0.3210	-0.5572	-3.83
4.415	-0.3204	-0.5577	-3.83
4.420	-0.3203	-0.5578	-3.83
4.425	-0.3197	-0.5578	-3.84
4.430	-0.3192	-0.5581	-3.84
4.435	-0.3191	-0.5581	-3.84
4.440	-0.3189	-0.5582	-3.84
4.445	-0.3188	-0.5584	-3.84
4.450	-0.3187	-0.5583	-3.84
4.455	-0.3186	-0.5589	-3.83
4.460	-0.3181	-0.5592	-3.83
4.465	-0.3178	-0.5597	-3.83
4.470	-0.3173	-0.5604	-3.82
4.475	-0.3169	-0.5612	-3.82
4.480	-0.3165	-0.5623	-3.81
4.485	-0.3161	-0.5632	-3.80

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.490	-0.3157	-0.5644	-3.79
4.495	-0.3151	-0.5653	-3.78
4.500	-0.3147	-0.5664	-3.77
4.505	-0.3143	-0.5671	-3.76
4.510	-0.3135	-0.5675	-3.76
4.515	-0.3132	-0.5678	-3.76
4.520	-0.3122	-0.5679	-3.77
4.525	-0.3119	-0.5678	-3.77
4.530	-0.3115	-0.5675	-3.78
4.535	-0.3113	-0.5673	-3.78
4.540	-0.3112	-0.5665	-3.79
4.545	-0.3111	-0.5658	-3.80
4.550	-0.3114	-0.5656	-3.80
4.555	-0.3113	-0.5652	-3.81
4.560	-0.3115	-0.5646	-3.81
4.565	-0.3115	-0.5642	-3.82
4.570	-0.3116	-0.5638	-3.82
4.575	-0.3119	-0.5629	-3.83
4.580	-0.3118	-0.5624	-3.83
4.585	-0.3123	-0.5618	-3.84
4.590	-0.3123	-0.5611	-3.85
4.595	-0.3126	-0.5604	-3.85
4.600	-0.3132	-0.5597	-3.86
4.605	-0.3133	-0.5590	-3.87
4.610	-0.3135	-0.5585	-3.87
4.615	-0.3140	-0.5577	-3.88
4.620	-0.3144	-0.5568	-3.88
4.625	-0.3145	-0.5561	-3.89
4.630	-0.3149	-0.5557	-3.89
4.635	-0.3150	-0.5548	-3.90
4.640	-0.3151	-0.5542	-3.91
4.645	-0.3154	-0.5535	-3.92
4.650	-0.3156	-0.5529	-3.92
4.655	-0.3155	-0.5520	-3.93
4.660	-0.3155	-0.5517	-3.94
4.665	-0.3155	-0.5511	-3.94
4.670	-0.3155	-0.5506	-3.95
4.675	-0.3152	-0.5500	-3.96
4.680	-0.3152	-0.5496	-3.96
4.685	-0.3149	-0.5492	-3.97
4.690	-0.3149	-0.5484	-3.98
4.695	-0.3148	-0.5481	-3.98
4.700	-0.3146	-0.5477	-3.99
4.705	-0.3143	-0.5471	-4.00
4.710	-0.3148	-0.5464	-4.01
4.715	-0.3149	-0.5455	-4.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	-0.3151	-0.5446	-4.02
4.725	-0.3157	-0.5437	-4.03
4.730	-0.3161	-0.5430	-4.04
4.735	-0.3165	-0.5417	-4.05
4.740	-0.3172	-0.5405	-4.06
4.745	-0.3179	-0.5391	-4.07
4.750	-0.3189	-0.5383	-4.07
4.755	-0.3197	-0.5373	-4.08
4.760	-0.3203	-0.5363	-4.09
4.765	-0.3209	-0.5356	-4.09
4.770	-0.3213	-0.5347	-4.10
4.775	-0.3215	-0.5343	-4.10
4.780	-0.3215	-0.5334	-4.11
4.785	-0.3216	-0.5333	-4.11
4.790	-0.3212	-0.5321	-4.13
4.795	-0.3205	-0.5318	-4.14
4.800	-0.3201	-0.5311	-4.15
4.805	-0.3197	-0.5307	-4.16
4.810	-0.3190	-0.5298	-4.17
4.815	-0.3184	-0.5291	-4.19
4.820	-0.3178	-0.5288	-4.19
4.825	-0.3170	-0.5283	-4.21
4.830	-0.3171	-0.5280	-4.21
4.835	-0.3169	-0.5278	-4.21
4.840	-0.3167	-0.5276	-4.22
4.845	-0.3168	-0.5277	-4.22
4.850	-0.3166	-0.5274	-4.22
4.855	-0.3168	-0.5280	-4.21
4.860	-0.3171	-0.5278	-4.21
4.865	-0.3172	-0.5277	-4.21
4.870	-0.3175	-0.5281	-4.21
4.875	-0.3179	-0.5285	-4.20
4.880	-0.3182	-0.5287	-4.19
4.885	-0.3184	-0.5290	-4.19
4.890	-0.3194	-0.5291	-4.18
4.895	-0.3193	-0.5299	-4.17
4.900	-0.3195	-0.5301	-4.17
4.905	-0.3199	-0.5303	-4.16
4.910	-0.3201	-0.5306	-4.16
4.915	-0.3206	-0.5307	-4.15
4.920	-0.3211	-0.5310	-4.15
4.925	-0.3213	-0.5312	-4.14
4.930	-0.3216	-0.5314	-4.14
4.935	-0.3220	-0.5317	-4.13
4.940	-0.3224	-0.5318	-4.13
4.945	-0.3228	-0.5320	-4.12

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	-0.3231	-0.5323	-4.11
4.955	-0.3236	-0.5323	-4.11
4.960	-0.3239	-0.5325	-4.11
4.965	-0.3243	-0.5326	-4.10
4.970	-0.3246	-0.5327	-4.10
4.975	-0.3250	-0.5327	-4.10
4.980	-0.3255	-0.5328	-4.09
4.985	-0.3256	-0.5327	-4.09
4.990	-0.3259	-0.5328	-4.09
4.995	-0.3260	-0.5329	-4.09
5.000	-0.3263	-0.5329	-4.08
5.005	-0.3263	-0.5330	-4.08
5.010	-0.3264	-0.5329	-4.08
5.015	-0.3263	-0.5331	-4.08
5.020	-0.3263	-0.5328	-4.08
5.025	-0.3260	-0.5332	-4.08
5.030	-0.3260	-0.5330	-4.09
5.035	-0.3257	-0.5335	-4.08
5.040	-0.3255	-0.5331	-4.09
5.045	-0.3251	-0.5338	-4.08
5.050	-0.3249	-0.5336	-4.09
5.055	-0.3247	-0.5340	-4.08
5.060	-0.3244	-0.5337	-4.09
5.065	-0.3247	-0.5345	-4.08
5.070	-0.3244	-0.5342	-4.08
5.075	-0.3249	-0.5341	-4.08
5.080	-0.3248	-0.5340	-4.08
5.085	-0.3256	-0.5337	-4.08
5.090	-0.3256	-0.5338	-4.08
5.095	-0.3268	-0.5333	-4.08
5.100	-0.3268	-0.5329	-4.08
5.105	-0.3279	-0.5323	-4.08
5.110	-0.3281	-0.5322	-4.08
5.115	-0.3286	-0.5317	-4.08
5.120	-0.3292	-0.5314	-4.08
5.125	-0.3299	-0.5308	-4.08
5.130	-0.3306	-0.5302	-4.08
5.135	-0.3311	-0.5298	-4.09
5.140	-0.3316	-0.5289	-4.09
5.145	-0.3321	-0.5284	-4.10
5.150	-0.3328	-0.5275	-4.10
5.155	-0.3336	-0.5264	-4.11
5.160	-0.3344	-0.5255	-4.11
5.165	-0.3350	-0.5245	-4.12
5.170	-0.3359	-0.5231	-4.13
5.175	-0.3369	-0.5221	-4.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.180	-0.3376	-0.5210	-4.14
5.185	-0.3383	-0.5198	-4.15
5.190	-0.3391	-0.5186	-4.16
5.195	-0.3400	-0.5169	-4.17
5.200	-0.3406	-0.5154	-4.18
5.205	-0.3417	-0.5139	-4.19
5.210	-0.3425	-0.5126	-4.20
5.215	-0.3432	-0.5110	-4.21
5.220	-0.3438	-0.5095	-4.23
5.225	-0.3447	-0.5080	-4.24
5.230	-0.3455	-0.5062	-4.25
5.235	-0.3463	-0.5046	-4.26
5.240	-0.3472	-0.5030	-4.28
5.245	-0.3478	-0.5014	-4.29
5.250	-0.3488	-0.4998	-4.30
5.255	-0.3497	-0.4981	-4.31
5.260	-0.3505	-0.4966	-4.32
5.265	-0.3514	-0.4950	-4.34
5.270	-0.3525	-0.4935	-4.34
5.275	-0.3532	-0.4920	-4.36
5.280	-0.3540	-0.4903	-4.37
5.285	-0.3550	-0.4889	-4.38
5.290	-0.3556	-0.4877	-4.39
5.295	-0.3567	-0.4863	-4.39
5.300	-0.3574	-0.4851	-4.40
5.305	-0.3584	-0.4840	-4.40
5.310	-0.3591	-0.4824	-4.42
5.315	-0.3600	-0.4813	-4.42
5.320	-0.3604	-0.4800	-4.43
5.325	-0.3611	-0.4787	-4.44
5.330	-0.3620	-0.4775	-4.45
5.335	-0.3624	-0.4761	-4.46
5.340	-0.3630	-0.4749	-4.47
5.345	-0.3635	-0.4738	-4.48
5.350	-0.3643	-0.4725	-4.49
5.355	-0.3648	-0.4711	-4.50
5.360	-0.3655	-0.4699	-4.51
5.365	-0.3663	-0.4684	-4.51
5.370	-0.3669	-0.4671	-4.52
5.375	-0.3676	-0.4658	-4.53
5.380	-0.3682	-0.4645	-4.54
5.385	-0.3691	-0.4630	-4.55
5.390	-0.3702	-0.4619	-4.55
5.395	-0.3711	-0.4607	-4.56
5.400	-0.3719	-0.4597	-4.56
5.405	-0.3733	-0.4588	-4.56

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.3741	-0.4577	-4.57
5.415	-0.3749	-0.4567	-4.57
5.420	-0.3758	-0.4559	-4.57
5.425	-0.3766	-0.4552	-4.57
5.430	-0.3775	-0.4542	-4.57
5.435	-0.3785	-0.4536	-4.57
5.440	-0.3791	-0.4528	-4.57
5.445	-0.3800	-0.4520	-4.58
5.450	-0.3807	-0.4514	-4.58
5.455	-0.3815	-0.4513	-4.57
5.460	-0.3820	-0.4502	-4.58
5.465	-0.3827	-0.4497	-4.58
5.470	-0.3834	-0.4490	-4.58
5.475	-0.3839	-0.4483	-4.58
5.480	-0.3847	-0.4477	-4.58
5.485	-0.3856	-0.4469	-4.58
5.490	-0.3863	-0.4463	-4.58
5.495	-0.3872	-0.4458	-4.58
5.500	-0.3881	-0.4452	-4.57
5.505	-0.3889	-0.4448	-4.57
5.510	-0.3900	-0.4445	-4.56
5.515	-0.3908	-0.4443	-4.56
5.520	-0.3917	-0.4444	-4.55
5.525	-0.3925	-0.4445	-4.54
5.530	-0.3934	-0.4444	-4.53
5.535	-0.3939	-0.4444	-4.53
5.540	-0.3948	-0.4446	-4.52
5.545	-0.3954	-0.4448	-4.51
5.550	-0.3960	-0.4448	-4.50
5.555	-0.3965	-0.4450	-4.49
5.560	-0.3974	-0.4450	-4.49
5.565	-0.3980	-0.4450	-4.48
5.570	-0.3986	-0.4451	-4.47
5.575	-0.3994	-0.4452	-4.46
5.580	-0.4000	-0.4451	-4.46
5.585	-0.4008	-0.4452	-4.45
5.590	-0.4018	-0.4453	-4.44
5.595	-0.4027	-0.4453	-4.43
5.600	-0.4036	-0.4454	-4.42
5.605	-0.4046	-0.4457	-4.41
5.610	-0.4054	-0.4459	-4.40
5.615	-0.4064	-0.4463	-4.39
5.620	-0.4071	-0.4465	-4.38
5.625	-0.4079	-0.4471	-4.36
5.630	-0.4085	-0.4473	-4.35
5.635	-0.4090	-0.4477	-4.35

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.4096	-0.4479	-4.34
5.645	-0.4102	-0.4484	-4.33
5.650	-0.4106	-0.4486	-4.32
5.655	-0.4110	-0.4488	-4.31
5.660	-0.4116	-0.4490	-4.31
5.665	-0.4120	-0.4492	-4.30
5.670	-0.4124	-0.4493	-4.30
5.675	-0.4130	-0.4496	-4.29
5.680	-0.4135	-0.4495	-4.28
5.685	-0.4139	-0.4497	-4.28
5.690	-0.4141	-0.4495	-4.28
5.695	-0.4144	-0.4498	-4.27
5.700	-0.4145	-0.4497	-4.27
5.705	-0.4149	-0.4497	-4.27
5.710	-0.4152	-0.4496	-4.26
5.715	-0.4152	-0.4500	-4.26
5.720	-0.4156	-0.4496	-4.26
5.725	-0.4156	-0.4499	-4.26
5.730	-0.4161	-0.4498	-4.25
5.735	-0.4160	-0.4498	-4.26
5.740	-0.4161	-0.4494	-4.26
5.745	-0.4163	-0.4490	-4.26
5.750	-0.4165	-0.4489	-4.26
5.755	-0.4165	-0.4488	-4.26
5.760	-0.4165	-0.4483	-4.27
5.765	-0.4169	-0.4481	-4.27
5.770	-0.4169	-0.4473	-4.27
5.775	-0.4171	-0.4470	-4.27
5.780	-0.4166	-0.4462	-4.29
5.785	-0.4172	-0.4458	-4.29
5.790	-0.4167	-0.4452	-4.30
5.795	-0.4169	-0.4442	-4.31
5.800	-0.4168	-0.4435	-4.31
5.805	-0.4169	-0.4426	-4.32
5.810	-0.4168	-0.4420	-4.33
5.815	-0.4166	-0.4409	-4.34
5.820	-0.4168	-0.4401	-4.35
5.825	-0.4170	-0.4388	-4.36
5.830	-0.4168	-0.4382	-4.37
5.835	-0.4170	-0.4369	-4.38
5.840	-0.4169	-0.4361	-4.39
5.845	-0.4170	-0.4349	-4.40
5.850	-0.4172	-0.4338	-4.41
5.855	-0.4172	-0.4329	-4.42
5.860	-0.4175	-0.4318	-4.43
5.865	-0.4178	-0.4307	-4.44

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.4179	-0.4297	-4.45
5.875	-0.4184	-0.4285	-4.45
5.880	-0.4186	-0.4276	-4.46
5.885	-0.4190	-0.4267	-4.47
5.890	-0.4192	-0.4258	-4.47
5.895	-0.4194	-0.4250	-4.48
5.900	-0.4198	-0.4243	-4.48
5.905	-0.4199	-0.4235	-4.49
5.910	-0.4203	-0.4227	-4.49
5.915	-0.4204	-0.4220	-4.50
5.920	-0.4205	-0.4212	-4.51
5.925	-0.4205	-0.4203	-4.52
5.930	-0.4205	-0.4196	-4.52
5.935	-0.4207	-0.4186	-4.53
5.940	-0.4209	-0.4179	-4.54
5.945	-0.4213	-0.4172	-4.54
5.950	-0.4213	-0.4161	-4.55
5.955	-0.4217	-0.4152	-4.56
5.960	-0.4221	-0.4144	-4.56
5.965	-0.4225	-0.4137	-4.56
5.970	-0.4233	-0.4128	-4.56
5.975	-0.4241	-0.4123	-4.56
5.980	-0.4246	-0.4119	-4.56
5.985	-0.4254	-0.4118	-4.55
5.990	-0.4257	-0.4117	-4.55
5.995	-0.4265	-0.4116	-4.54
6.000	-0.4269	-0.4117	-4.54

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable (|S11_ref1-S11_ref2| < 1e-02). Consider starting the sweep above the affected range.

- 1 frequency point(s) with $\varepsilon'' < 0$ (non-passive, worst ε'' = -0.452), attributable to measurement noise on a near-lossless liquid. Values kept and flagged, not replaced by another root.